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Graduation Project I

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Abstract

Masarak is a Saudi-based Smart Travel & Lifestyle Super-App that integrates all essential day-to-day services and travel into one digital platform. With this app, ride service providers, food delivery companies, hotel booking services and grocery stores are brought together in one mobile app, thereby addressing the fragmented services currently experienced by tourists and citizens living in the Kingdom of Saudi Arabia.

The digital transformation of the nation depends heavily on this enterprise, along with the growth of tourism and plans for developing smart cities in line with Vision 2030. The platform utilizes a machine-learning recommendation engine that uses data of the users' location and their general behavior, and it will give users personalized experience by providing specific recommendations when using the service (e.g., where the user is located).

Our app has a mobile front end; secure RESTful APIs; and a database to keep track of all user data as well as ensuring the platform can accommodate large numbers of users while providing them with high availability and scalable performance. To protect both end users accounts and financial transactions from fraud, we encrypt communication through TLS; authenticate users through the use of JWTs (JSON Web Tokens); and restrict access to user and financial transactions based on roles.

The goals of this project are to help reduce the fragmentation of services through combining many core service categories into one platform, thereby resulting in greater efficiencies for users through providing them with personalized service recommendations and with secure, unified access to all three core service areas. The architecture is designed to be modular and extensible so that it can be expanded in the future to incorporate additional services.

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Chapter 1:

INTRODUCTION

1.1 Background of the Project:

Saudi Arabia has been undergoing rapid digital transformation as per its Vision 2030 national strategy. This has been predominantly driven by key goals such as economic diversification to improve living standards, expansion of tourism through mega-projects like NEOM, the Red Sea Project and the urban development of Riyadh and smart digital services provided to end-users.

While there are many digital services available to users in Saudi Arabia today, users still heavily rely on many different apps on a daily basis for things such as travel (Uber), food delivery (Hunger Station), hotel booking services (Booking), and grocery shopping (Ninja). Due to the number of apps that users are using, users will often find themselves frustrated by several inefficiencies; re-entering their name, email, telephone number and home address each time they use an app; having to create new accounts because they can't remember which app/email they used to register; or having different levels of service based on which app they used.

To help solve these problems, the concept for Masarak, is to create a Saudi-based Smart Travel & Lifestyle Super-App that will integrate the most important apps that users need into one platform. The Masarak Super-App will provide users with an AI powered seamless experience that will help provide users with tailored experiences both for tourists and residents of KSA.

1.2 Problem Statement

As of now users in the Kingdom of Saudi Arabia are facing:

- Service fragmentation between apps (multiple rides, food and market apps).
- Time lost switching between ride/food/market apps.
- Different prices between apps and you can't compare them in real time.
- Little to no personalization from one app to another.
- Redundant data between apps and multiple accounts for payments.

It is crucial to create an integrated intelligent platform to connect all of these required services into one single digital ecosystem where everything works seamlessly together.

1.3 Objectives of the Project:

- To achieve the overall goal of building and developing a secure scalable Smart Travel & Living Super App (Masarak); the specific goals will be: Develop an integrated App that enables users to book rides, hotels, order food, and purchase groceries from local providers.
- Incorporate advanced features and deployment, including AI-based personalized recommendations, cloud hosting for global access, and an administrative dashboard for system management.

1.4 Significance of the Project:

The project helps Saudi Arabia achieve its digital transformation goals through the creation of a Unified Smart Travel and Lifestyle Super-App which combines essential services into a single intelligent platform. The system provides users with greater convenience through

improved service access which supports national strategic objectives including Vision 2030 and the FIFA World Cup 2034.

1.4.1 Benefits for Tourists

- One Platform: The application enables users to access food delivery services together with transportation and hotel and flight and grocery and event services through a single application.
- Real-Time Comparison: The system enables users to compare prices together with product availability and customer ratings through immediate access to information which helps them make better choices.
- AI-Based Recommendations: The system delivers customized travel and lifestyle recommendations which depend on user preferences together with their current geographical location.
- Time Saving: The system enables users to complete their booking process and find services within five minutes after they previously needed 20 to 30 minutes.
- World Cup Readiness: The FIFA World Cup 2034 event together with all international events will offer users simplified options for navigation and accommodation and transportation and event discovery.
- Multilingual Support: The interface provides Arabic and English language options which meet the needs of international users.

1.4.2 Benefits for Residents

- Daily Convenience: Users can manage their transportation needs together with their food requirements and shopping activities and entertainment options through a single unified system.
- Cost Optimization: The system enables users to discover optimal deals by conducting instant price assessments.
- Smart Lifestyle Insights: The system uses AI technology to create usage-based recommendations for users.
- Reduced App Fragmentation: Users need to install and handle fewer applications which results in less application management work.

The system enables users to control all their discounts and loyalty programs and seasonal promotional events through one central platform.

1.3.3 Benefits for Service Providers & Businesses

- Expanded Market Access: The system provides access to both local citizens and foreign visitors through its unified platform.
- Higher Visibility: The system uses artificial intelligence to deliver exposure which matches user demand and their particular interests.
- Analytics & Insights: The system provides access to both booking patterns and user interaction data together with performance measurement tools.
- Operational Efficiency: The system requires only one main integration platform which eliminates the need for various independent partnership connections.

- Increased Revenue Potential: The service demand during peak tourism periods and key events like the World Cup will increase by 20 to 30 percent.

1.3.4 Benefits for Saudi Arabia (Vision 2030 Alignment)

- Supports Vision 2030 Goals: The project supports three tourism goals throughout its digital economy development work and its efforts to develop the private sector.
- Enhances Global Image: The system establishes Saudi Arabia as an advanced smart tourism destination.
- World Cup Infrastructure Support: The system enables visitors to experience FIFA World Cup 2034 through a smooth and uninterrupted visiting process.
- Economic Impact: The study shows that people will spend money throughout the hospitality industry and the transportation system and the retail sector and the entertainment industry.
- Data-Driven Planning: The system collects data from multiple sources to support urban development and smart city projects and tourism management activities.

1.5 Scope of the Project:

Masarak is a Saudi-based Smart Travel & Lifestyle Super-App designed to integrate essential daily and travel-related services into one unified digital platform.

1.5.1. In Scope

1. Development of three core service modules:

- Ride service
- Hotel booking services
- Food delivery services
- Market and grocery delivery services

2. Implementation of user account management features, including:
    - User registration and secure login
    - Profile creation and modification
3. Integration of service interaction features, including:
    - Browsing available services
    - Real-time comparison of service options
    - Booking services and purchasing products
    - Tracking the status of service requests
4. Development of an AI-based recommendation system to provide:
    - Personalized suggestions
    - Location-based recommendations
    - Behavior-based service recommendations
5. Development of the system's technical infrastructure, including:
    - Mobile application (frontend)
    - Backend API services
    - Database design and management
    - User authentication and authorization mechanisms
    - Payment integration using a sandbox environment
    - Administrative dashboard

- System testing for performance, usability, and reliability

1.5.2 Out of Scope

- Ticket booking for FIFA WORLD CUP 2034 matches.
- Event reservation services.
- Flight booking integration.
- Full commercial integration with live third-party service providers.
- Advanced AI predictive analytics beyond the basic recommendation system.
- International deployment outside Saudi Arabia.
- Full production-level payment gateway implementation.

1.6 Limitations of the Project

1. Academic Time Constraints

The development schedule of Masarak needs to be restricted because it functions as an educational project. The development team will begin their work on advanced system components after completing the initial stage of the project.

2. Prototype-Level Implementation

The system is currently in the prototype development phase and will not function as a fully commercial super-application during its first stage.

3. Sandbox Payment and Simulated APIs

The application uses sandbox payment gateways and simulated APIs instead of real vendors and service providers. The system will function differently from a commercial platform that operates at full capacity.

4. Limited AI Training Data

The AI-based recommendation system requires strong technical infrastructure and access to large, high-quality datasets. The testing process needs real user data but testing shows to produce less accurate results with limited information available.

5. Third-Party Integration Constraints

All live service provider integrations need official partnerships and API access permissions which need to be obtained through proper channels. The implementation of real-time integration will experience delays because of legal and technical and operational requirements in the initial stage.

6. Infrastructure and Resource Limitations

The system development process will face scalability and performance challenges because of cloud resources and computational power and server capacity restrictions which exist during the early development stages.

1.7. Project Plan

The phases of the project will consist of the following:

- Phase 1 - Project Proposal (February 1-17, 2026)
- Phase 2 - Chapter 1: Introduction (February 18 - March 9, 2026)
- Phase 3 - Chapter 2: Literature Review (March 10 - April 6, 2026)
- Phase 4 - Chapter 3: System Analysis & Design (April 7 - April 20, 2026)
- Phase 5 - First Draft Submission (April 21-May 4, 2026)
- Phase 6 - Final Draft Submission (May 5-11, 2026)
- Final Defense (May 17-20, 2026)

The above phases of the project are to occur in accordance with respective official submission deadlines, allowing for enough time to complete documentation, receive supervisor feedback, and revise work as necessary.

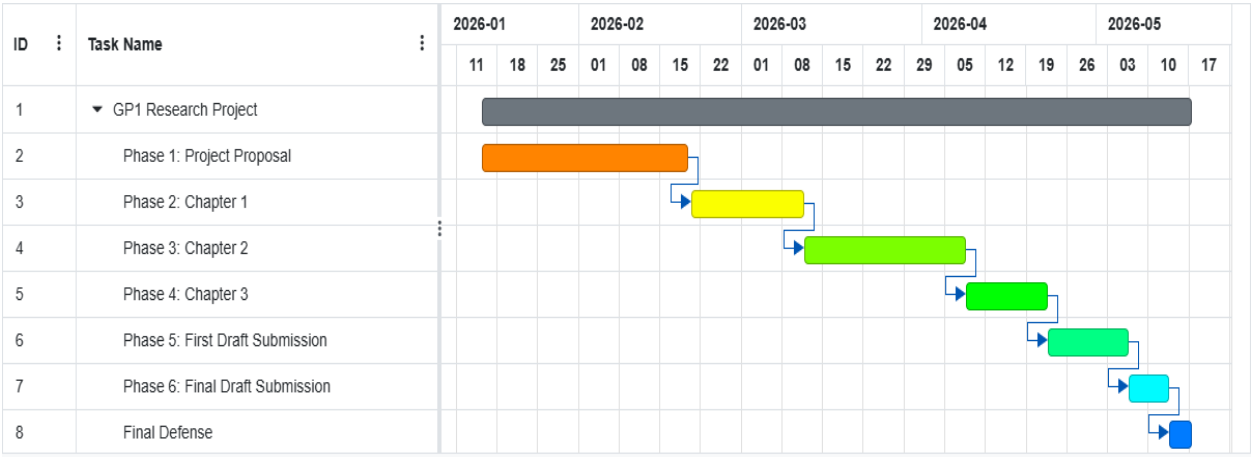


Figure 1: *Gantt chart*

1.8 Literature Review

As digital transformation initiatives continue growing tremendously throughout the world, the rapid emergence of a new type of integrated "super-app" ecosystem is emerging. A super-app is an integrated digital platform that provides access to several related services (e.g., transportation, delivery, payment) in one application

(TechTarget, n.d.). Through this aggregation of services there will be less disjointed service delivery across the related areas while providing greater organizational efficiency by centralizing user management and enabling ease of access to all services from one location.

The two examples of Grab and Careem illustrate this type of model. For instance, Grab has evolved from offering only ride-hailing to a super-app that includes food delivery, digital payments, financial services and logistics to users throughout Southeast Asia (Grab Holdings Limited, 2023). Careem has entered the Middle East and is now delivering ride-hailing, delivery and digital wallet services within its own ecosystem (Careem, 2023). These two examples demonstrate how the integration of services can improve customer loyalty and increase the capabilities of the platform.

From a technology perspective, most of today's super apps use microservice architecture and an API-based connect model. Microservice architecture helps improve how scalable, modular, and maintainable a distributed system is (Newman, 2015).

Development teams have a modular and independent method to supply service modules for RESTful APIs, making it easy to achieve interoperability between the various service modules as well as providing developers with the ability to enhance or add upon the functionality of an individual service module.

AI provides a vast number of possibilities for improving the user experience. There are many ways an individual can engage with applications on a super app platform. Research has established the correlation between collaborative filtering and behavioral personalization in providing satisfactory service to users and in determining whether services provided to users meet their expectations, as stated by Ricci, Rokach, and Shapira (2015). The use of historical user data, user behavior, and spatial data allows for improved efficiency of service delivery through personalized services based on historical user data, spatial data, and user behavior.

Cybersecurity remains a significant concern when developing an integrated digital ecosystem. Stallings (2018) identifies the importance of applying secure authentication methods, building a secure communication channel through the use of TLS and/or SSL, and following RBAC (Role-Based Access Control) procedures to address unauthorized access and data breaches for cloud-based applications. Several developers are now employing the use of JSON Web Tokens (JWT) as a token-based authentication framework to secure their application programming interfaces (APIs) in a distributed environment.

In the case of Saudi Arabia, Vision 2030 is an initiative to increase digital transformation, develop smart cities and diversify the economy (Vision 2030, n.d.). With an increasing reliance on digital services to assist with transportation, delivery and e-commerce, there is a clear opportunity to develop and implement integrated platforms that help to eliminate redundancy and improve efficiency as well.

The literature reviewed supports the architectural and strategic design of Masarak by emphasizing:

- Integrated service ecosystems (Super-App model)
- AI-based personalization
- Microservices and API-based architecture
- Secure authentication and encrypted communications
- Cloud-based scalability

Overall, this research confirms that Masarak is a viable and relevant scalable Smart Travel & Lifestyle Super-App within the context of Saudi Arabia's digital transformation.

1.9 Conclusion

Masarak functions as a Smart Travel and Lifestyle Super-App which combines ride services and food delivery and grocery services into a single digital platform. The system reduces service fragmentation because it enables users to experience a complete and safe and customized service which operates through artificial intelligence and cloud-based service solutions. Masarak developed its system to handle future growth while it supports Saudi Vision 2030 through its digital transformation efforts and creation of smart connected services.

Chapter 2:

LITERATURE REVIEW

2.1 Introduction

In this chapter, we compared our proposed system to other similar existing systems based on the functionalities and features, highlighting both the strengths and weaknesses of each system. This has provided valuable insight that helped inform design decisions in later chapters. The following table sums up the literature review in this section.

2.2 Booking

Booking.com operates as an online travel booking platform which allows users to find and evaluate and book various travel services while its primary service provides hotel reservation services. The platform supports a wide range of properties, including hotels, apartments, and resorts, which makes it suitable for different types of travelers.

The application provides several core functionalities that enhance user experience. Users can search for accommodation based on location and travel dates, they can apply advanced filters which include price and ratings and amenities and property type, and they can view detailed listings which include images and descriptions and verified customer reviews. The platform enables users to make bookings and handle their reservations while they obtain current information about available rooms and pricing. The system uses secure authentication methods together with payment processing technologies to handle these activities.

Booking.com provides a user-friendly platform which includes an interface that supports multiple languages and multiple currencies while enabling users to book flights and car

rentals. The platform introduces basic tailored recommendations which help users discover suitable options while enhancing their overall experience.

The platform has several strengths that contribute to its success, including its extensive global coverage, large variety of accommodation options, reliable review system, and efficient real-time booking capabilities. These factors enhance user trust and make the platform convenient and accessible for a wide range of users.

Because of its various constraints, the platform lacks a broad range of effective functionality. The scope of the platform is travel-related services; however, travel-related services do not include any daily life-style services, such as ride-hailing or food delivery. Because of differences in the pricing structure across various platforms, there are large variances in the prices displayed in the platform itself. In addition, the platform as a whole and many of the features provided are dependent upon third-party providers, which can lead to inconsistency in service levels and availability.

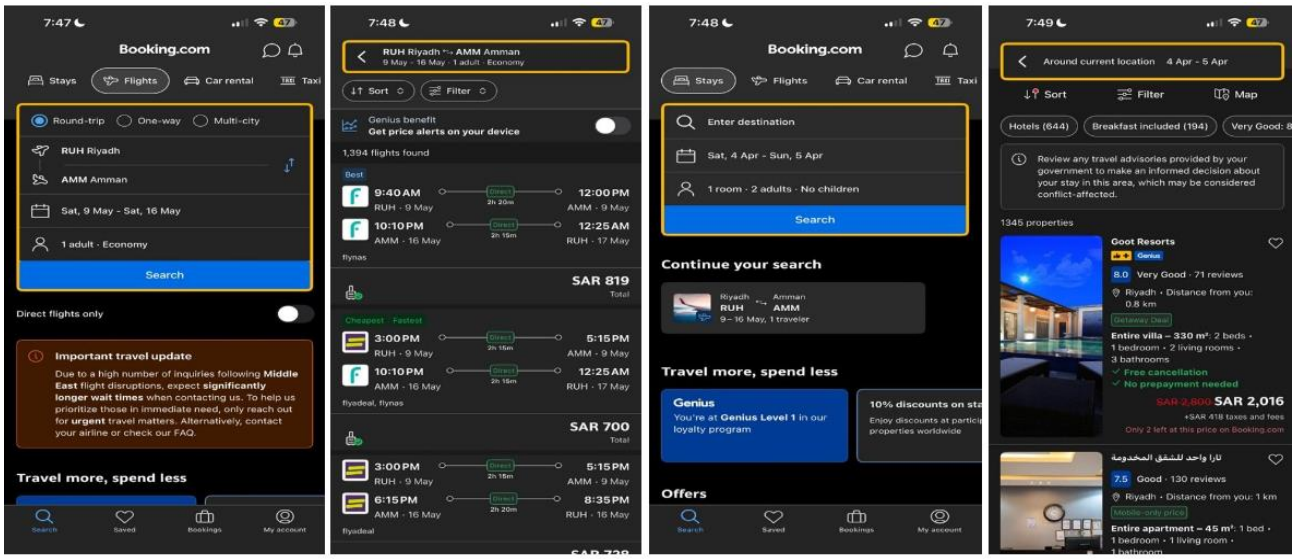


Figure 2 :*Booking.com Application*

2.3 Careem

Careem feels like one of the clearest examples of a regional super-app done seriously. The application has expanded beyond its original purpose of providing ride-hailing services. Careem presents itself as "the everything app" of the greater Middle East because it enables users to handle multiple functions from one application. The company states that its services have reached more than 50 million customers who use its platform across 70 cities in 10 different countries. This widespread operation in multiple countries establishes the company as an important standard for evaluating this specific type of platform.

The system achieves its highest performance level through its single platform which provides all needed functions. The value is simple: one app, many daily needs. Careem

Plus provides users with savings across multiple services which include rides, food, groceries, and dining, whereas Careem Pay enables users to pay bills and send money. The interesting aspect of Careem's AI applications for personalization and local relevance research stems from their plan to develop food recommendation systems based on user patterns and their method to enhance ETAs through local weather conditions and prayer schedules and Ramadan iftar times. The company demonstrates its operational capabilities through its partnership with Tamimi Markets, which operates in Riyadh, Saudi Arabia, to create a successful delivery system through its local partner network, which provides fast grocery delivery service. Careem operates with actual operational limitations.

The concept of an "everything app" appears to provide complete service, yet the actual services delivered throughout different locations exhibit varying levels of service provision. Careem's help pages state that service availability differentials exist across cities and markets, while different booking options become available to users based on their geographical area. The situation becomes more difficult because multiple systems depend on each other for their operation. The terms of Careem explicitly indicate that various application features of the platform come from outside partners instead of the company itself.

The application experience presents itself as one cohesive system, but different aspects of the system will operate differently based on regional factors and available resources and local laws. Careem simplifies human life by uniting multiple needs into one common application which serves as its core strength.

Its weakness is that a super-app is only as strong as the cities, partners, and services behind it. That is what makes Careem impressive, but also what makes the model hard to perfect.

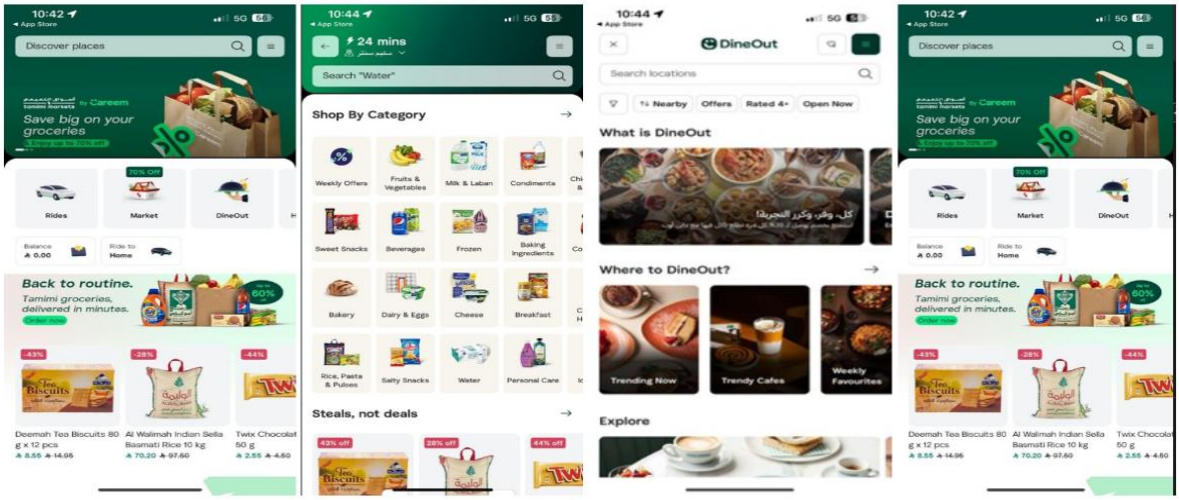


Figure 3: *Careem.com Application*

2.4 Grab

Grab Holdings is a multi-faceted digital platform that provides an all-in-one solution for transportation, delivery, and financial services. It's a versatile tool which enables customers to book rides (cars, taxis, or motorbikes), order food through GrabFood, and send packages/parcels via delivery services. It can meet virtually all of your daily needs!

Grab's app includes many capabilities to make it easier for customers to use the service. A customer can book a ride with Grab based on where they are at the time of booking and where they want to go. They can also order food from restaurants and grocery items from stores. Lastly, they can send packages using Grab as a delivery service. The app allows customers to pay with a credit card (using GrabPay) and track their orders/rides in real-

time using GPS tracking systems, an effective authentication system (to secure their accounts), and integrated payment systems.

The Grab app is easy to navigate. Users, along with drivers and partners who are delivering packages, can easily communicate during the app's use. Promotion opportunities and a rewards program have been added to promote continued use of the Grab app. Also, users can find their booking history, receive digital receipts, and get personalized and relevant recommendations when using the Grab app.

The Grab platform and app provide positive results to users (vis-à-vis other app providers) because the Grab platform can provide multiple types of services all within one app; it has a broad reach across Southeast Asia, has access to many drivers and merchants to provide service for customers, and it has a fast real-time tracking service for an efficient use of the new app. All of these features and functions provide users with an easier and more accessible way to utilize the Grab app.

There are some drawbacks that may impact how users feel about their experience with Grab. For example, these include: the application is significantly more expensive to use at peak times due to surge pricing, in addition there may be added service fees on top of what are already incurred by the service when completing their transactions, the quality of services provided through drivers and merchants can vary widely from one user to another, users may experience delays or cancellations and the Grab mobile application can also put

a lot of stress on device hardware and battery usage, and response times from the customer service department may not be uniform.

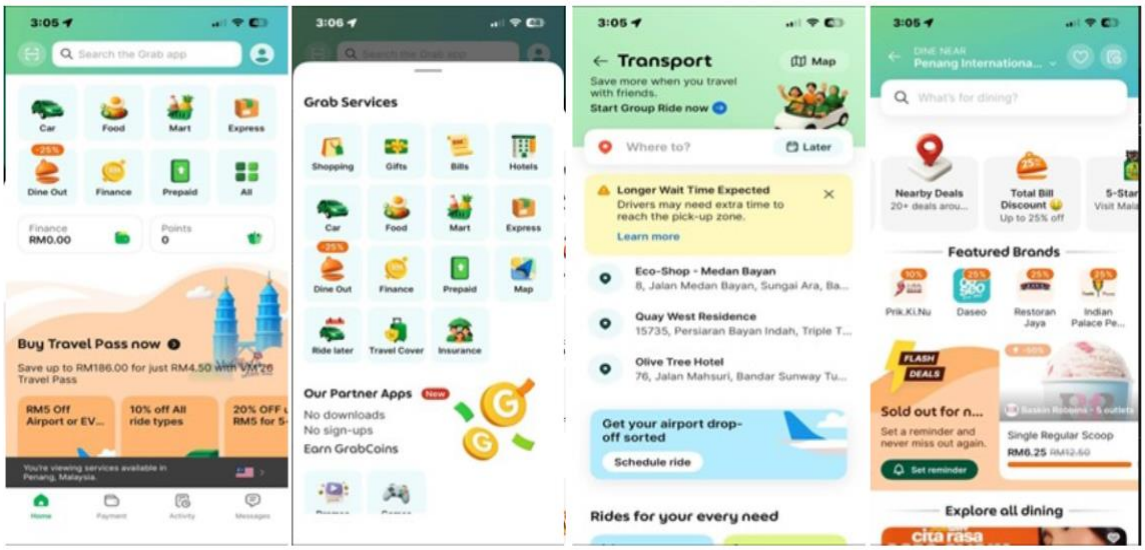


***Figure 4**: Grab Application*

2.5 Comparison

Feature	Booking.com	Careem	Grab	Masarak
Multi-Service Integration	✗	✓	✓	✓
Travel Booking (Flights/Hotels)	✓	✗	✗	✓
Ride Hailing Services	✗	✓	✓	✓
Food & Grocery Delivery	✗	✓	✓	✓
AI Powered Personalization	Basic	Advanced	Advanced	Highly Advanced
Smart Trip Planning	✗	✗	✗	✓
Unified Login / Single Account	✓	✓	✓	✓
Need for multiple Apps	✓	✗	✗	✗
Real-Time Tracking (Ride/ Delivery)	✗	✓	✓	✓

Real-Time Optimization (Time/Cost/Route)	✗	✓	✓	✓
Saudi Arabia Focus	✗	✓	✗	✓
Cultural & Local Adaption	✗	✓	Limited	✓

Table 1: *Literature Review*

Modern digital platform comparison table lists key characteristics. Multi-service integration means users can use multiple services from one application. The user experience is improved with AI driven personalization and smart trip planning resulting in personalized recommendations and easier bookings. Features like single sign in reduce the necessity of multiple accounts. The unified login makes it easy to find your way around and to track your trip in real time. Furthermore, focusing on and adjusting for the Saudi Arabian culture will provide users with a more relevant and effective overall experience.

2.6 Conclusion

Research of current platforms, like Booking.com, Careem, and Grab show all have a solid foundation but don't provide a total user experience. Booking.com is considered the best source for travel and lodging but does not contain a connection to items that are used on a daily basis. On the other hand, Careem and Grab do provide the benefit of a super-app by having the ride-share and food delivery combined but struggle due to their geographic limitations and needing a partnership to utilize their services.

These factors show that there is an opening for a product in Saudi Arabia, where people are using multiple applications to complete a task, creating fragmentation and inefficiency in their daily routines. This is affected by the current Vision 2030 of Saudi Arabia and calls to develop a smarter digital Saudi Arabia.

Masarak intends to solve these issues and offers a super-app using AI, that's been developed specifically for Saudi Arabia. Masarak integrates the travel, transportation and daily lifestyle apps into a unified platform, thereby offering greater usability to the customer. In addition, by decreasing the need for multiple applications to complete a task, Masarak allows for the user to develop a personalized and localized experience. Ultimately, Masarak provides a greater level of cohesion and readiness to move forward with the Kingdom's expectations of a digitally smart Saudi Arabia.

Chapter 3:

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

The chapter demonstrates the design work and total system architecture, which develops the Masarak application. The system design document provides an in-depth description of its main system elements, which include the mobile application (frontend), backend services, database, and AI recommendation module. The chapter provides details about the system's component interactions, which enable data movement throughout the system to maintain operational efficiency and dependable performance. The section examines essential design factors that include scalability, security, and performance, which create a seamless experience for users.

3.2 Survey Analysis

The survey conducted is to analyze the users' behavior, preferences, and challenges with the existing service applications. The results obtained by the survey provided the key data needed for the design process of the Masarak application.

3.2.1 Overview

The purpose of this survey was to gain insight into user behavior, preferences and problems using current service applications such as ride-sharing services, food delivery, and grocery platforms. The survey aimed to measure the demand for an integrated super-application type of solution (i.e. Masarak) for users as well as to determine what features users want. Responses gathered from this survey will be used as a reliable basis for analyzing user needs and assisting in establishing system design decisions.

3.2.2 Demographic Analysis

The majority of survey participants were in the 18-24 age range (young adults), who have a tendency to use mobile apps quite frequently. Also, approximately 70% of all respondents were students, which indicates that they are a digital and tech-savvy group of users overall.

The participants also had academic backgrounds covering a wide variety of fields, with many from technology-related fields. With regard to gender distribution; approximately 77% of the respondents are female and the rest of the participants were male.

Therefore, the demographic information reflects users who are generally very engaged with digital based services and who will benefit from having a single, unified type of a super application.

3.2.3 Findings & Insights

According to the survey findings, service application users use at least 2-3 different services on a weekly or daily basis. The fact that many people who responded would likely to switch from one service to another was considered inconvenient and time-consuming points to a great need for a unified solution.

Users expressed great interest in having a single application that could provide several different types of services. Most users indicated they would use this application. The most important features identified by users were real-time pricing, AI-based recommendations, and real time tracking. This points to the demand for greater efficiency and personalization.

Numerous survey participants were concerned with security; therefore, majority of respondents placed significant value on securing their payment system and protecting their private information. The overall findings indicate that there is a potential for growth opportunities for Masarak due to these findings, providing insight into how to fulfil or exceed user expectations.

3.2.4 Charts

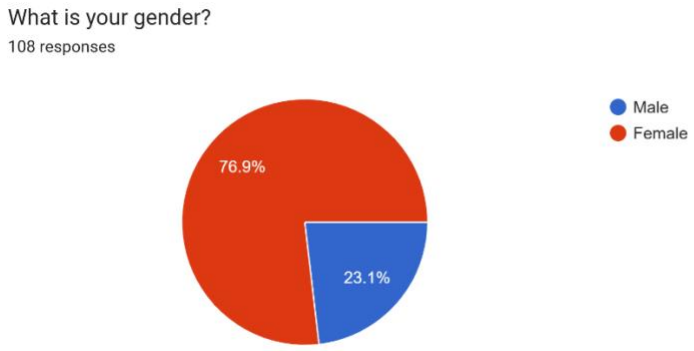


Figure 5 *: Gender Chart*

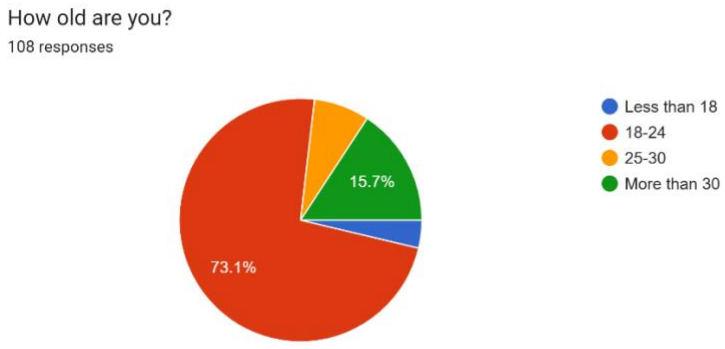


Figure 6: *Age Chart*

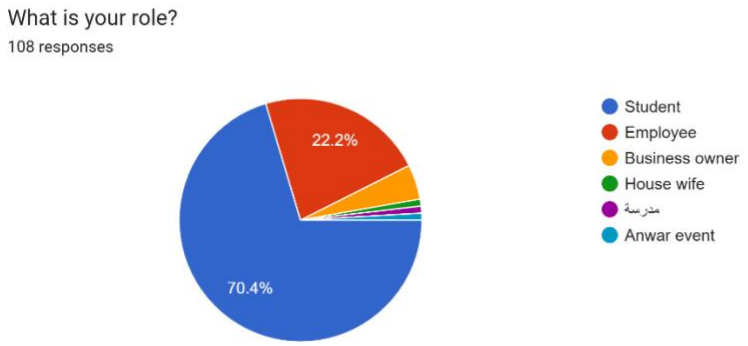


Figure 7 : *Role Chart*

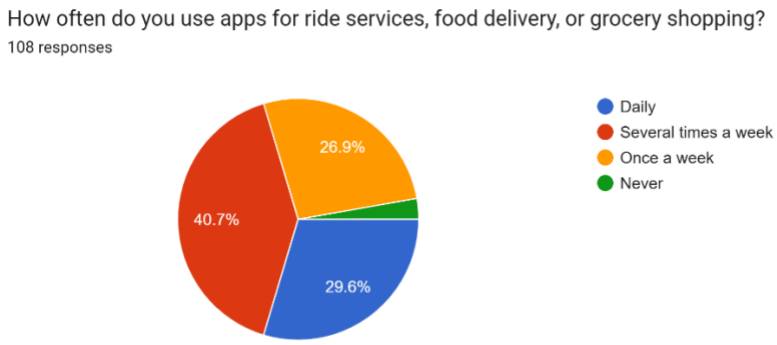


Figure 8: *App Usage Chart*

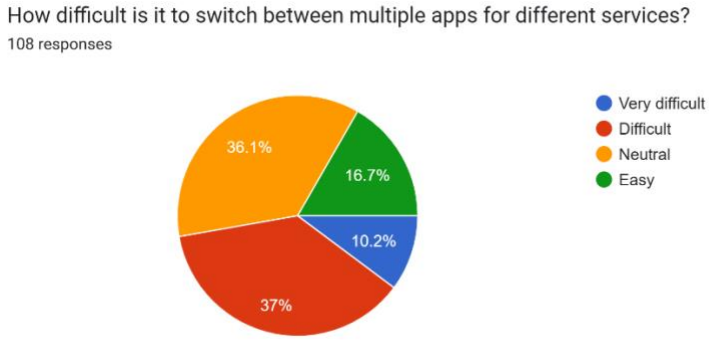


Figure 9 : *App Switching Chart*

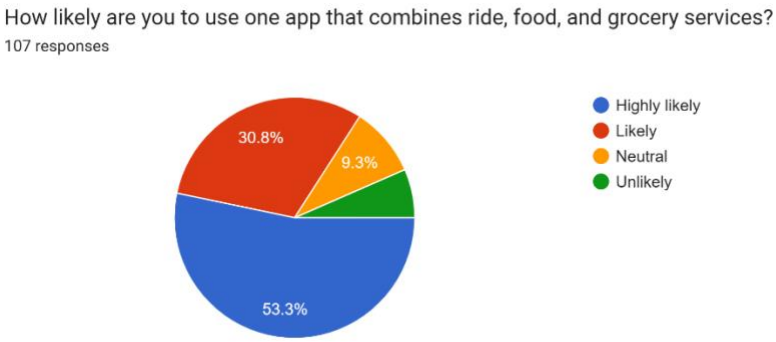


Figure 10: *Unified App Usage Chart*

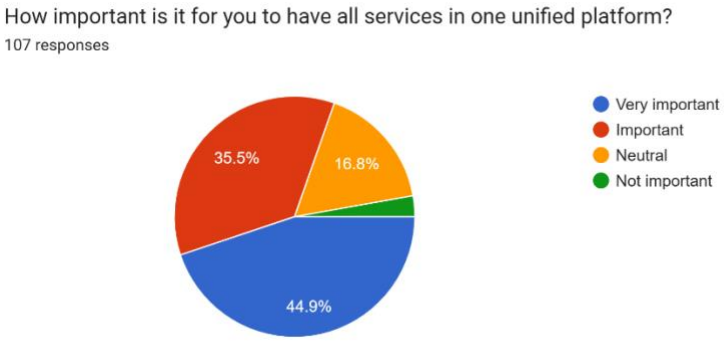


Figure 11: *Unified Platform Chart*

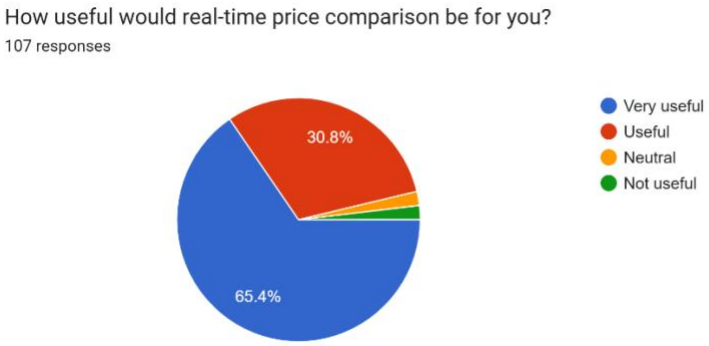


Figure 12*: Price Comparison Chart*

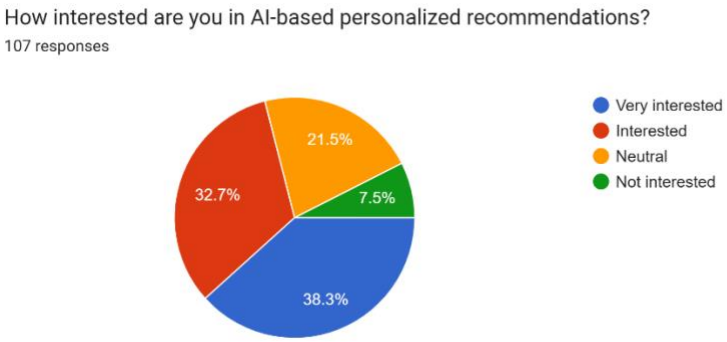


Figure 13: *AI Recommendation Chart*

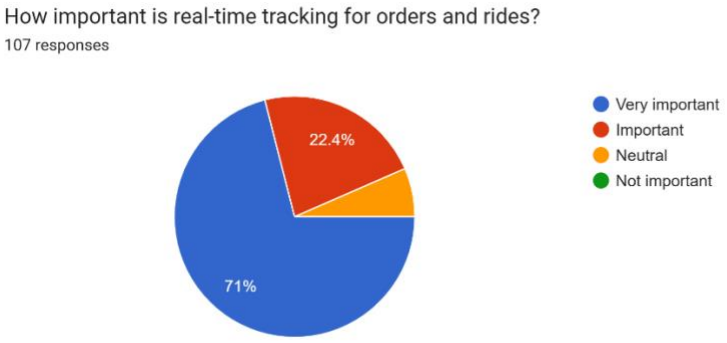


Figure 14 : *Real-Time Tracking Chart*

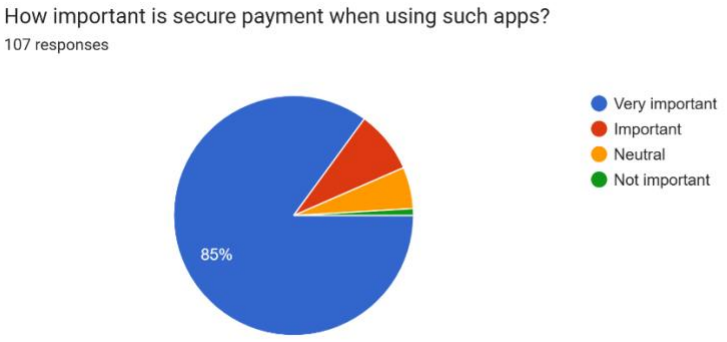


Figure 15: *Secure Payment Chart*

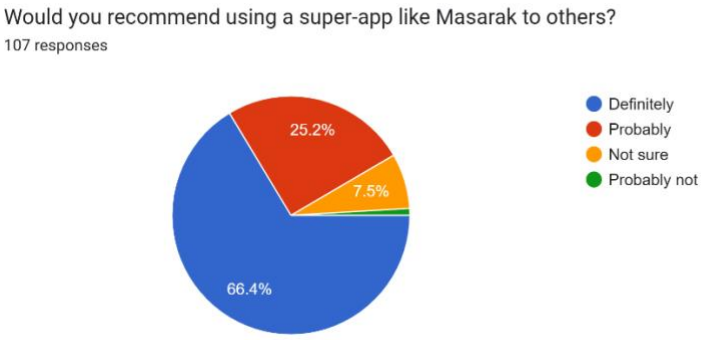


Figure 16 : *Recommendation Chart*

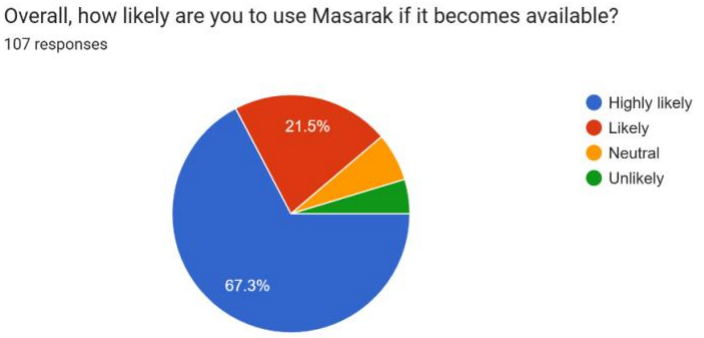


Figure 17*: Masarak Usage Chart*

3.2.5 Conclusion

According to survey results, a nearly unanimous amount of evidence shows the concept of having Masarak as an integrated application is possible and realistic. Respondents stated that they experience numerous difficulties and issues when utilizing multiple platforms to access different services, but would much rather have one solution offering their entire suite of services (as opposed to using multiple solutions).Security, real time tracking, and intelligent recommendations are the three most important features required to meet user expectations.

3.3 User Characteristics

As per the survey, Masarak's primary users are users of digital services who use multiple applications to get their daily tasks done on a regular basis.

These users are considered to be technologically adept, and they expect speed, reliability and ease of use from applications. They typically like simple and intuitive interfaces which allow them to perform their tasks quickly and easily with very little effort.

These users value the convenience of having many functions available to them; as they will frequently switch between multiple services (for example, transport service and food ordering service) therefore, demonstrating the necessity for a unified service that streamlines their experience.

These users also desire personalized features such as AI-based recommendations, as well as the functionality of being able to track the places and compare prices in real-time. Additional factors further contributed towards importance of security and data protection especially concerning payment-related services.

Due to their efficiency and convenience-driven users, Masarak is a perfect candidate for a unified super-app.

3.4 Requirement Engineering

Functional requirements:

- The system shall allow users to create an account using email or phone number.
- The system shall validate all user inputs during registration and login to ensure correctness and prevent malicious data.
- The system shall allow users to securely log in and log out of the application.
- The system shall authenticate users using secure authentication mechanisms and ensure that only authorized users can access their accounts.
- The system shall allow users to manage their personal profiles, including viewing and updating their information.
- The system shall allow users to access multiple services within a single application, including ride booking, food ordering, and grocery shopping.
- The system shall allow users to browse available services and view details such as price, ratings, and availability.
- The system shall allow users to compare prices between different service providers in real time.
- The system shall allow users to book rides by selecting pickup and destination locations.
- The system shall allow users to place food and grocery orders easily and efficiently.

- The system shall ensure that only authenticated users can perform booking and purchasing operations.
- The system shall allow users to track their rides and orders in real time.
- The system shall integrate with external service providers and validate all received data before processing.
- The system shall provide personalized recommendations based on user preferences, behavior, and location.
- The system shall analyze user interactions to improve recommendation accuracy over time.
- The system shall allow users to perform secure payment transactions within the application.
- The system shall encrypt all payment-related data during transmission.
- The system shall not store sensitive financial information such as full card details.
- The system shall store transaction records securely and allow users to view their history and receipts.
- The system shall allow administrators to manage users, services, and system data.
- The system shall restrict administrative operations to authorized roles only.
- The system shall support secure payment using 3D Secure authentication and OTP verification.

Non-functional requirements:

- The system shall be scalable to support future growth and additional services.
- The system shall provide a user-friendly and intuitive interface.
- The system shall support both Arabic and English languages.
- The system shall handle errors gracefully and display clear messages without exposing sensitive information.
- The system shall be modular and maintainable to support future updates and enhancements.
- The system shall ensure secure communication using HTTPS/TLS protocols.
- The system shall store sensitive data such as passwords using hashing or encryption techniques.
- The system shall monitor activities and log events to detect suspicious behavior.
- The system shall protect user privacy by collecting only necessary data and preventing unauthorized access or sharing.

Acceptance criteria:

- Users shall be able to successfully create an account and log in using valid credentials.
- Users shall receive an error message when incorrect login information is entered.
- Users shall be able to access multiple services within one application.
- Users shall be able to compare prices between service providers before making decisions.
- Users shall be able to successfully book rides and place orders.
- Users shall be able to track their rides and orders in real time.
- Users shall receive personalized recommendations based on their preferences.
- Users shall be able to complete secure payment transactions without exposing sensitive information.
- Unauthorized users shall not be able to access restricted features.
- The system shall respond quickly and operate without crashes during normal use.
- The system shall display clear and user-friendly error messages when issues occur.

3.5.4 Survey traceability matrix:

- Since the survey showed that users rely on multiple applications for different services, the system integrates ride booking, food ordering, and grocery shopping into one platform to improve convenience.
- Since users reported difficulty switching between applications, the system provides a unified interface that allows access to all services in one place.
- Since price comparison was identified as an important need, the system includes real-time price comparison between service providers.
- Since real-time tracking was highly requested by users, the system provides live tracking of rides and orders.
- Since users showed interest in personalized services, the system includes recommendation features based on user behavior and location.
- Since security and payment safety were major concerns, the system ensures secure authentication, encrypted communication, and safe payment processing.
- Since users expressed concerns about privacy, the system protects user data and limits unnecessary data collection.
- Since users expect fast and responsive applications, the system is designed to provide high performance and quick response time.
- Since ease of use was important to users, the system provides a simple and intuitive user interface.
- Since users require reliable services, the system ensures high availability and minimal downtime.

3.5 System Design

3.5.1 Use case diagram

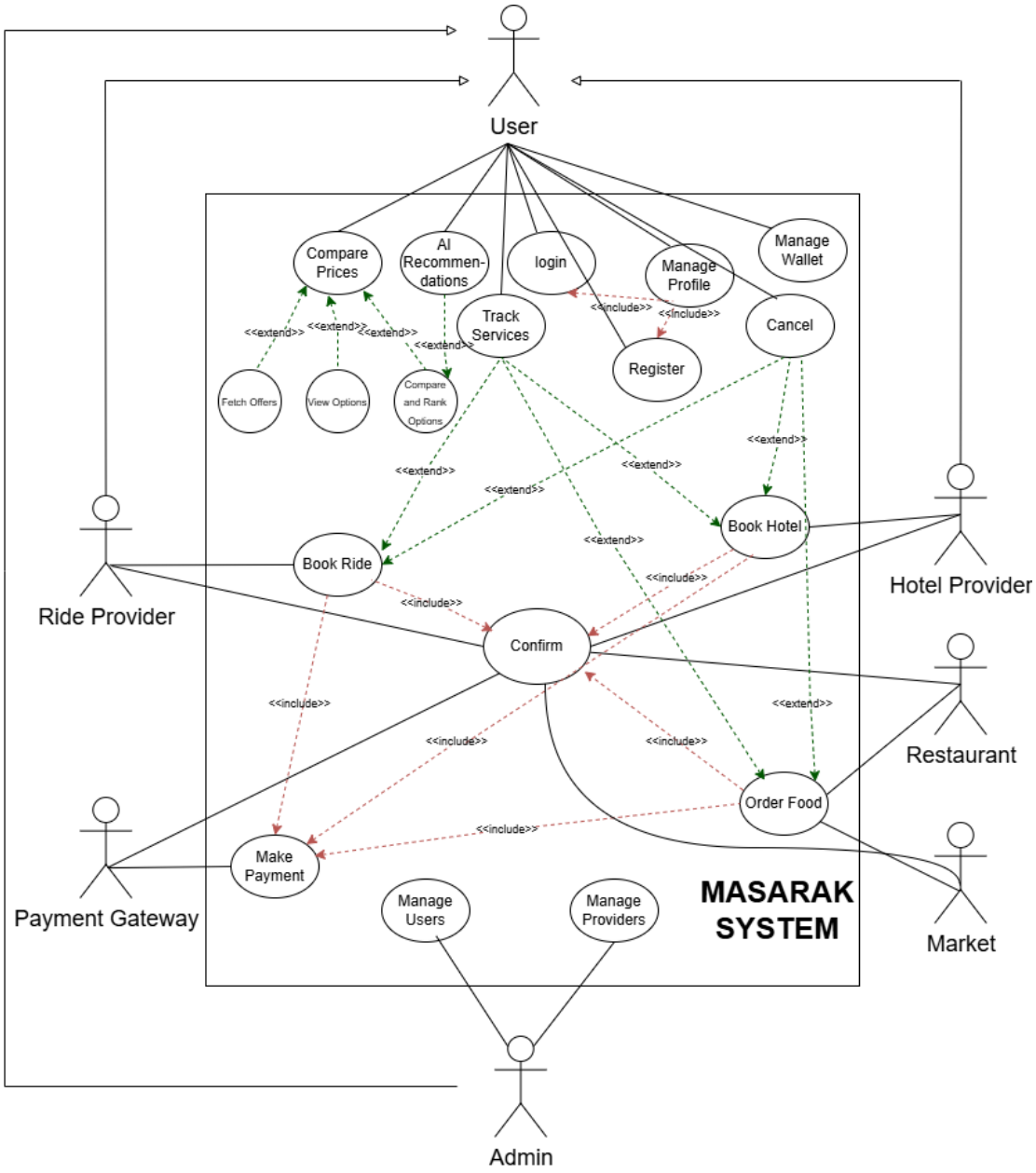


Figure 18: *Use case diagram*

3.5.2 Use case descriptions

<table>
<tr><td>Use Case ID</td><td colspan="2">UC-01</td></tr>
<tr><td>Use Case Name</td><td colspan="2">Register</td></tr>
<tr><td>Actors</td><td colspan="2">• Primary: User
• Supporting: System</td></tr>
<tr><td>Description</td><td colspan="2">Allows a new user to create an account in the Masarak System.</td></tr>
<tr><td>Preconditions</td><td colspan="2">• User is not registered</td></tr>
<tr><td>Main Flow</td><td>Actor Action</td><td>System response</td></tr>
<tr><td></td><td>1. User opens registration page

2. User enters personal data

3. User submits form</td><td>1. System displays registration.

2. System validates input

3. System creates account and confirms</td></tr>
<tr><td>Postconditions</td><td colspan="2">• Account is successfully created</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">• User cancels registration</td></tr>
<tr><td>Exception Flows</td><td colspan="2">• Email already exists → System shows error</td></tr>
<tr><td>Dependency</td><td colspan="2">• None</td></tr>
<tr><td>Assumptions</td><td colspan="2">• User provides valid data</td></tr>
</table>

Table 2: *Use Case Description 1: Register*

Use Case ID	**UC-02**	
Use Case Name	Login	
Actors	• Primary: User	
Description	Allows users to log into the system.	
Preconditions	• User is registered	
Main Flow	**Actor Action**	**System response**
	1. Enter credentials 2. Click login	1. System validates 2. System grants access
Postconditions	• User logged in successfully	
Alternative Flow	• Forgot password	
Exception Flows	• Invalid credentials	
Dependency	• Registration	
Assumptions	• User remembers credentials	

Table 3: *Use Case Description 2: Login*

<table>
<tr><td>Use Case ID</td><td>UC-03</td></tr>
<tr><td>Use Case Name</td><td>Compare Prices</td></tr>
<tr><td>Actors</td><td>• Primary: User
• Supporting: Providers</td></tr>
<tr><td>Description</td><td>Allows users to compare services (rides, hotels, food).</td></tr>
<tr><td>Preconditions</td><td>• User is logged in</td></tr>
<tr><td>Main Flow</td><td><table><tr><th>Actor Action</th><th>System response</th></tr><tr><td>1. Select service

2. Request comparison

3. View ranking</td><td>1. System fetches offers

2. System shows options

3 System ranks results</td></tr></table></td></tr>
<tr><td>Postconditions</td><td>• Comparison results displayed</td></tr>
<tr><td>Alternative Flow</td><td>• Apply filters</td></tr>
<tr><td>Exception Flows</td><td>• No data available</td></tr>
<tr><td>Dependency</td><td>• Fetch Offers, View Options</td></tr>
<tr><td>Assumptions</td><td>• Providers share updated data</td></tr>
</table>

Table 4: *Use Case Description 3: Compare Prices*

<table>
<tr><td>Use Case ID</td><td>UC-04</td></tr>
<tr><td>Use Case Name</td><td>Book Ride</td></tr>
<tr><td>Actors</td><td>• Primary: User
• Supporting: Ride Provider</td></tr>
<tr><td>Description</td><td>Allows users to book a ride.</td></tr>
<tr><td>Preconditions</td><td>• Logged in
• Ride available</td></tr>
<tr><td>Main Flow</td><td><table><tr><th>Actor Action</th><th>System response</th></tr><tr><td>1. Select ride

2. Confirm

3. Proceed</td><td>1. Show details.

2. Create booking

3. Payment</td></tr></table></td></tr>
<tr><td>Postconditions</td><td>• Ride booked</td></tr>
<tr><td>Alternative Flow</td><td>• Cancel booking</td></tr>
<tr><td>Exception Flows</td><td>• Ride unavailable</td></tr>
<tr><td>Dependency</td><td>• Compare Prices, Payment</td></tr>
<tr><td>Assumptions</td><td>• Provider is available</td></tr>
</table>

Table 5: *Use Case Description 4: Book Ride*

<table>
<tr><td>Use Case ID</td><td>UC-05</td></tr>
<tr><td>Use Case Name</td><td>Order Food</td></tr>
<tr><td>Actors</td><td>• Primary: User
• Supporting: Restaurant</td></tr>
<tr><td>Description</td><td>Allow users to order food.</td></tr>
<tr><td>Preconditions</td><td>• Logged in</td></tr>
<tr><td>Main Flow</td><td><table><tr><th>Actor Action</th><th>System response</th></tr><tr><td>1. Choose restaurant

2. Select items

3. Confirm</td><td>1. Show menu.

2. Update cart

3. Process order</td></tr></table></td></tr>
<tr><td>Postconditions</td><td>• Order placed</td></tr>
<tr><td>Alternative Flow</td><td>• Modify cart</td></tr>
<tr><td>Exception Flows</td><td>• Item unavailable</td></tr>
<tr><td>Dependency</td><td>• Payment</td></tr>
<tr><td>Assumptions</td><td>• Menu is accurate</td></tr>
</table>

Table 6: *Use Case Description 5: Order Food*

3.5.3 Sequence diagram

1. Login Sequence Diagram

This sequence diagram illustrates the login process in the Masarak application, including retry functionality. The user enters their email and password through the mobile application, which sends the credentials securely to the authentication service. The service validates the data against the database. If the credentials are valid, a token is generated and returned to the application for secure access. If the credentials are invalid or the user does not exist, an error message is displayed. A loop fragment is used to represent that the user can retry the login process multiple times until successful authentication is achieved.

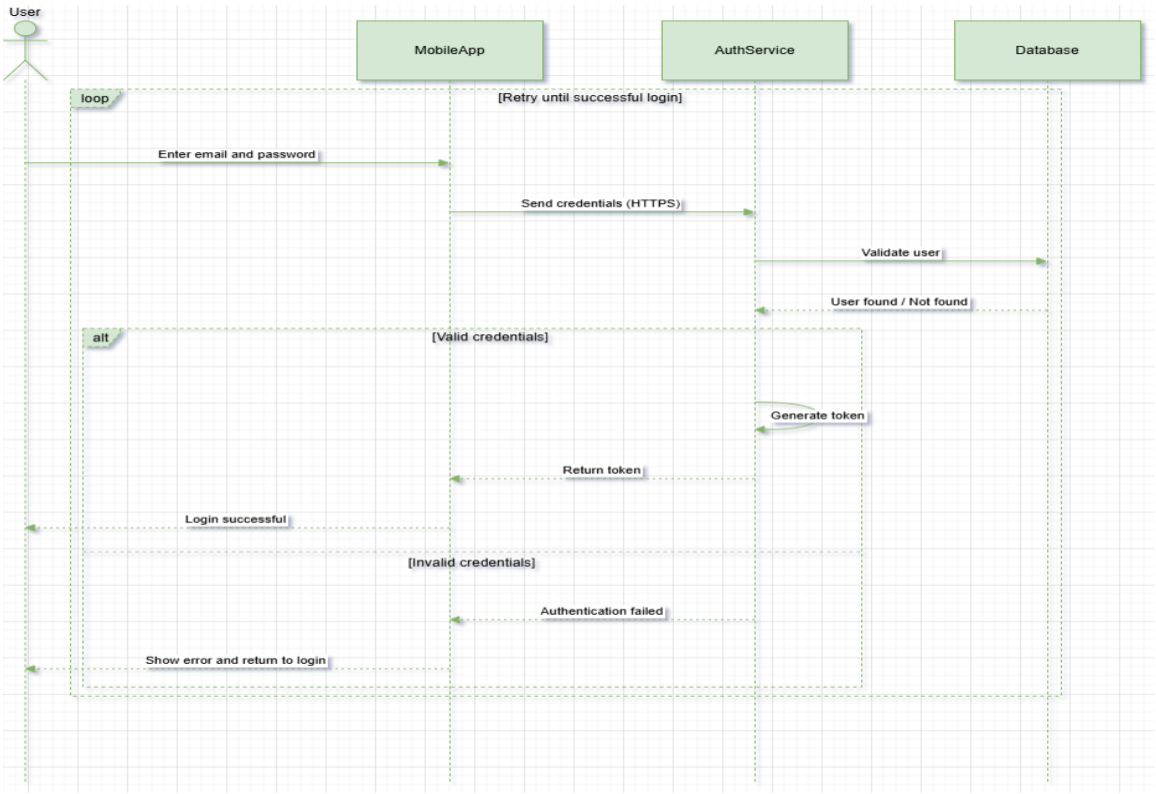


Figure 19 : *Login sequence diagram*

2. Ride Booking Sequence Diagram

This sequence diagram illustrates the process of booking a ride service with integrated provider comparison. The user submits a ride request through the mobile application, which forwards it to the backend system. The backend retrieves offers from multiple external providers (such as ride-hailing platforms) and compares them based on factors like price and availability. The ranked options are displayed to the user, who selects a preferred provider. The backend then checks availability and confirms the booking if possible. If no rides are available, the system notifies the user with an appropriate error message. This diagram demonstrates both decision-making and booking functionality within a unified flow.

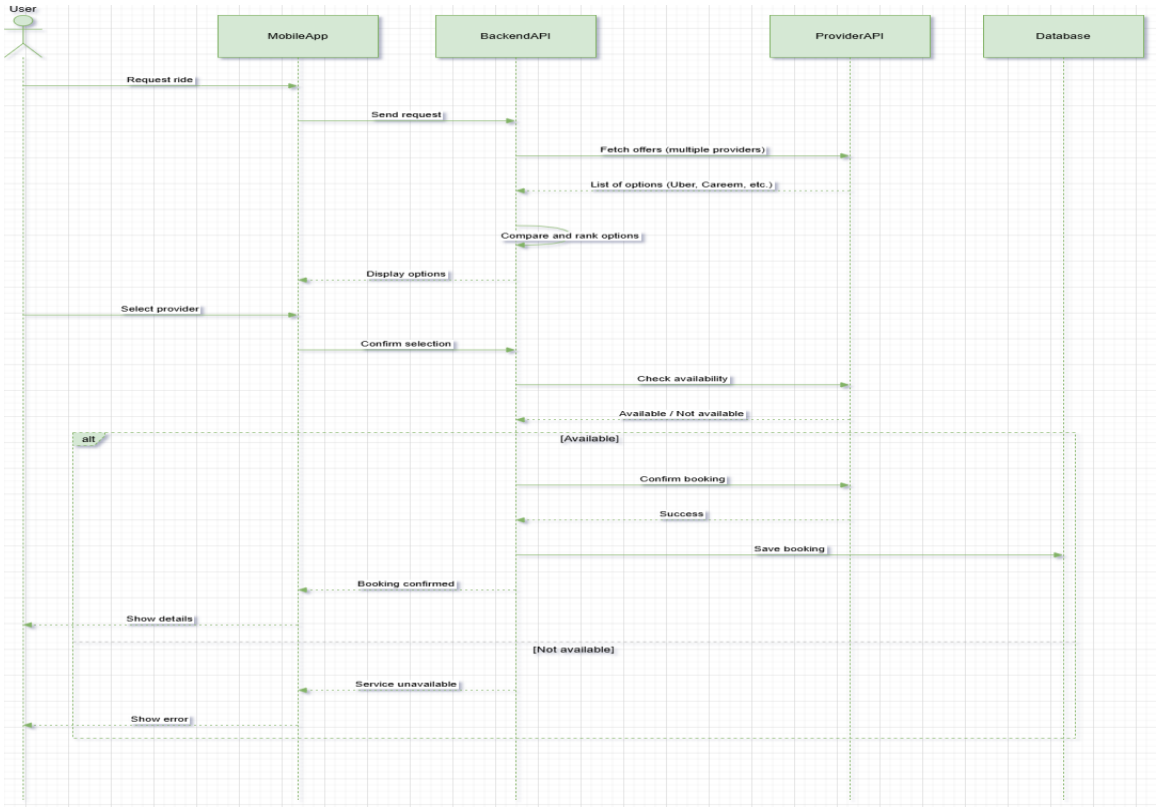


Figure 20: *Ride Booking Services Sequence Diagram*

3. Food Booking Sequence Diagram

This sequence diagram represents the process of ordering food with integrated comparison across multiple providers. The user browses food options and sends a request through the mobile application. The backend system gathers data from various food service providers, including menus, prices, and delivery times. It then compares and ranks the available options before presenting them to the user. After the user selects a preferred provider, the backend checks item availability and confirms the order if possible. If the requested items are not available, an error message is displayed. This diagram highlights efficient service comparison and reliable order handling.

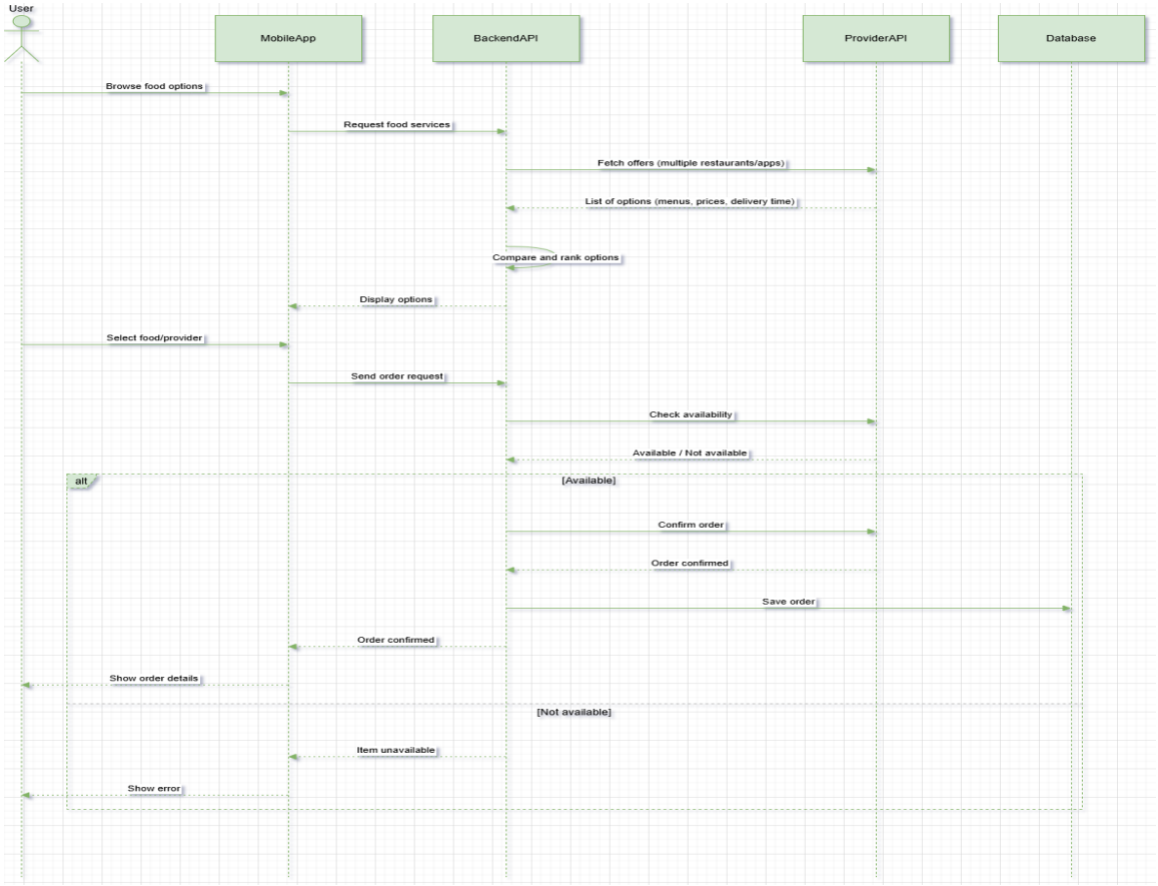


Figure 21: *Food Booking Sequence Diagram*

4. Market Booking Sequence Diagram

This diagram illustrates the process of ordering products from market services with integrated comparison functionality. The user searches for products via the mobile application, and the backend retrieves options from multiple stores. These options are compared based on factors such as price and availability, and the results are presented to the user. Once the user selects a product and provider, the backend verifies stock availability and proceeds with order confirmation if the item is in stock. If the product is unavailable, the system informs the user accordingly. This diagram demonstrates effective inventory-based service handling combined with comparison logic.

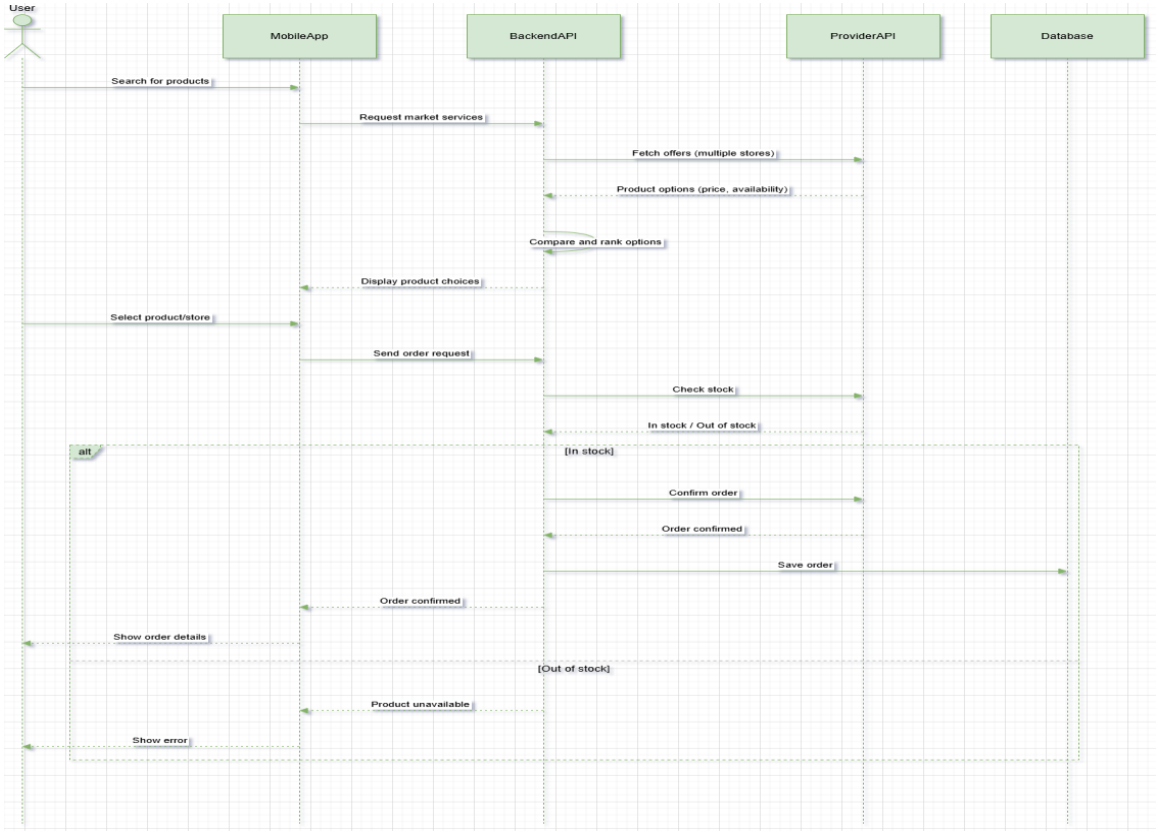


***Figure 22**: Market Booking Sequence Diagram*

5. Hotel Booking Sequence Diagram

This sequence diagram shows how a user books a hotel through the system. The request is sent from the mobile application to the backend, which verifies room availability through an external hotel API. If rooms are available, the booking is confirmed and stored in the database, and the user receives reservation details. If the hotel is fully booked, the system notifies the user accordingly. The diagram ensures realistic handling of availability constraints.

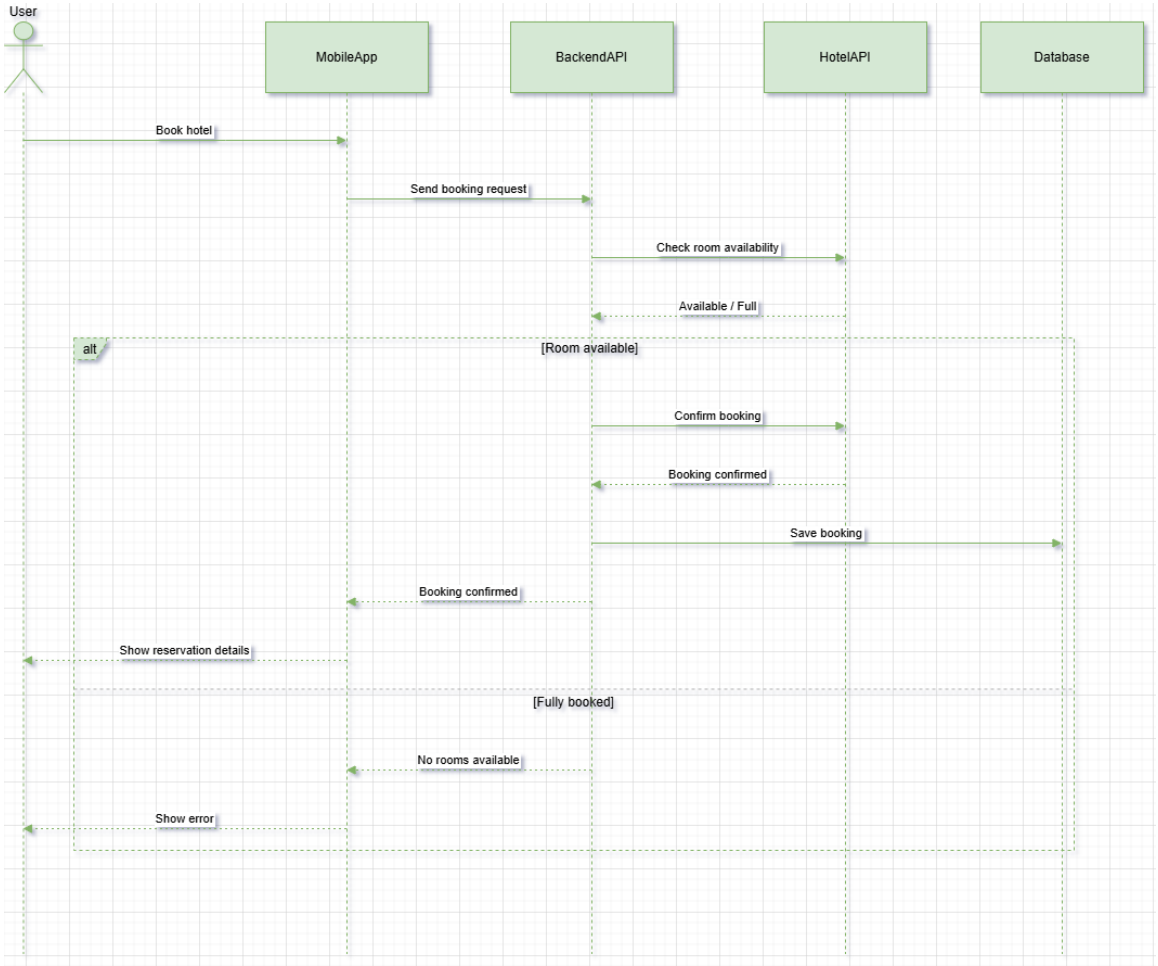


Figure 23: *Hotel Booking Sequence Diagram*

6. Payment Sequence Diagram

This sequence diagram illustrates the secure payment process in the Masarak system, supporting both wallet and card payment methods. The payment request is sent through HTTPS/TLS along with a user authentication token to ensure secure communication and access control. The backend system validates the token before processing the transaction. If the user selects wallet payment, the system checks the wallet balance and deducts the amount if sufficient funds are available. If the balance is insufficient, the payment fails. For card payment, the backend sends a tokenized card payment request to the payment gateway. The payment gateway communicates with the bank system to perform 3D Secure authentication using an OTP. The user enters the OTP, and the bank verifies both the OTP and the available balance before approving or declining the payment. The final payment status is securely updated in the database and displayed to the user.

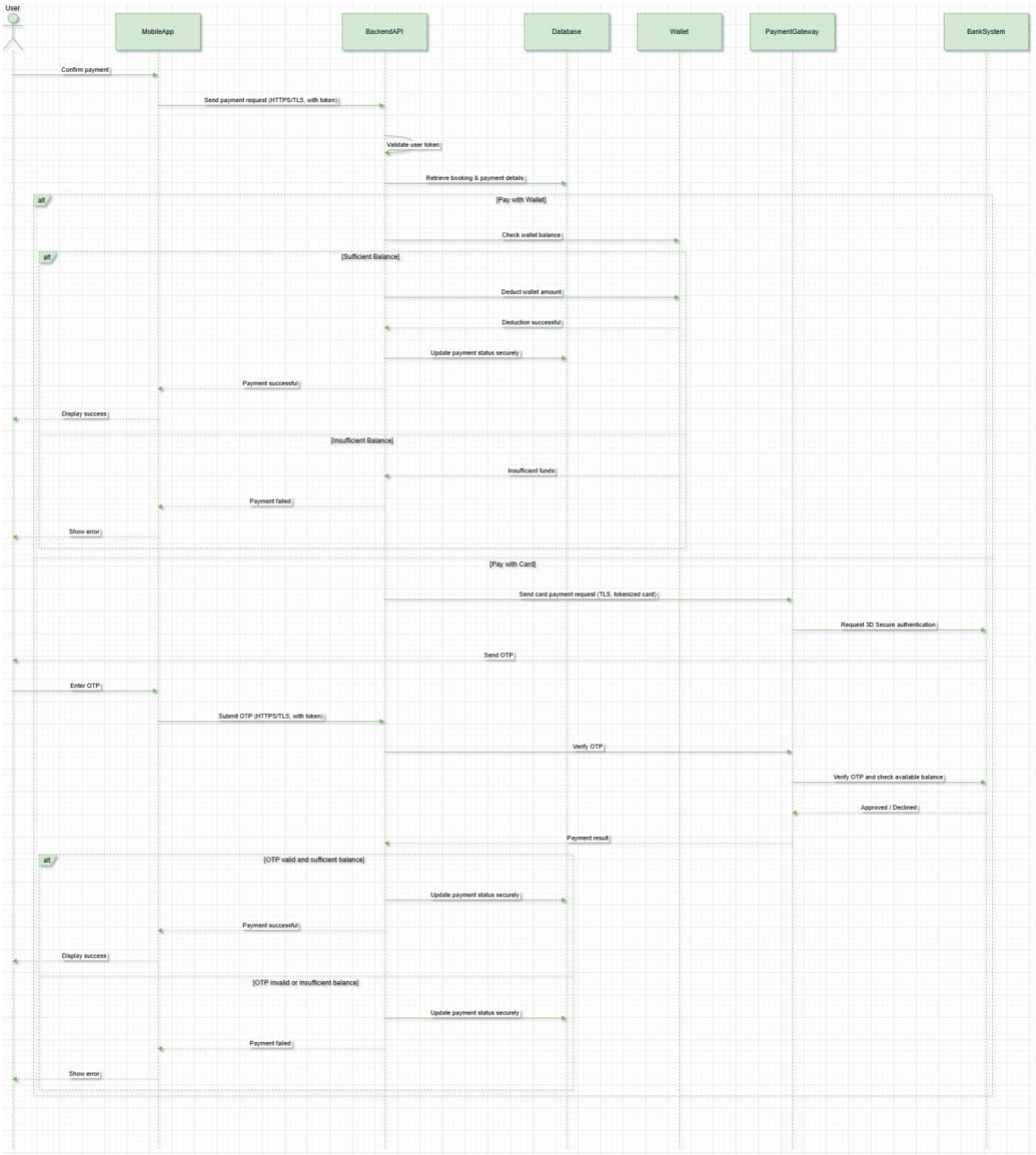


Figure 24: *Payment Sequence Diagram*

7. Wallet Sequence Diagram

This diagram illustrates how users can add funds to their wallet within the application. The user initiates a top-up request, which is processed through an external payment gateway. Upon successful payment, the wallet service updates the user's balance and stores the updated information in the database. The system then confirms the updated balance to the user. This process ensures seamless and secure wallet management.

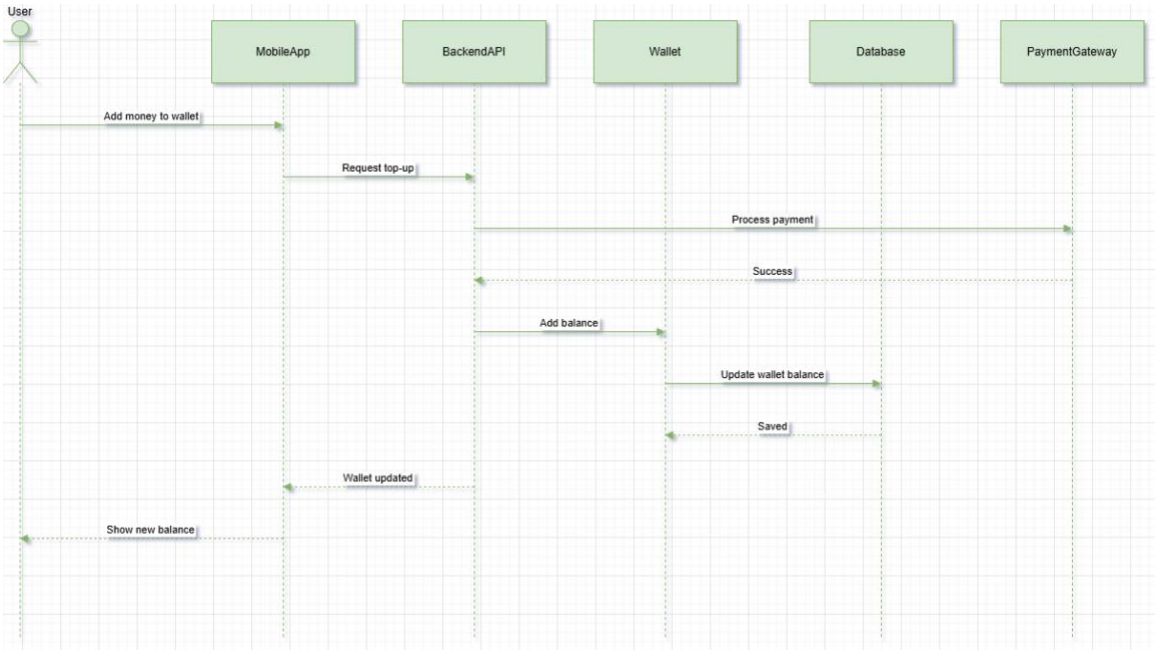


***Figure 25**: Wallet Sequence Diagram*

8. AI Assistant Sequence Diagram

This sequence diagram illustrates how the AI assistant generates personalized recommendations within the Masarak application. When the user requests suggestions, the mobile application sends the request to the backend system. The backend retrieves user-specific data from the database, including preferences, previous interactions, and contextual information such as location and time. This data is then forwarded to the AI engine, where it is analyzed using recommendation algorithms. The AI engine processes the input, ranks the available services, and generates a list of personalized suggestions. If sufficient user data is available, tailored recommendations are returned; otherwise, the system provides general suggestions. The final results are delivered to the user through the mobile application, enhancing the overall user experience through intelligent and adaptive decision-making.

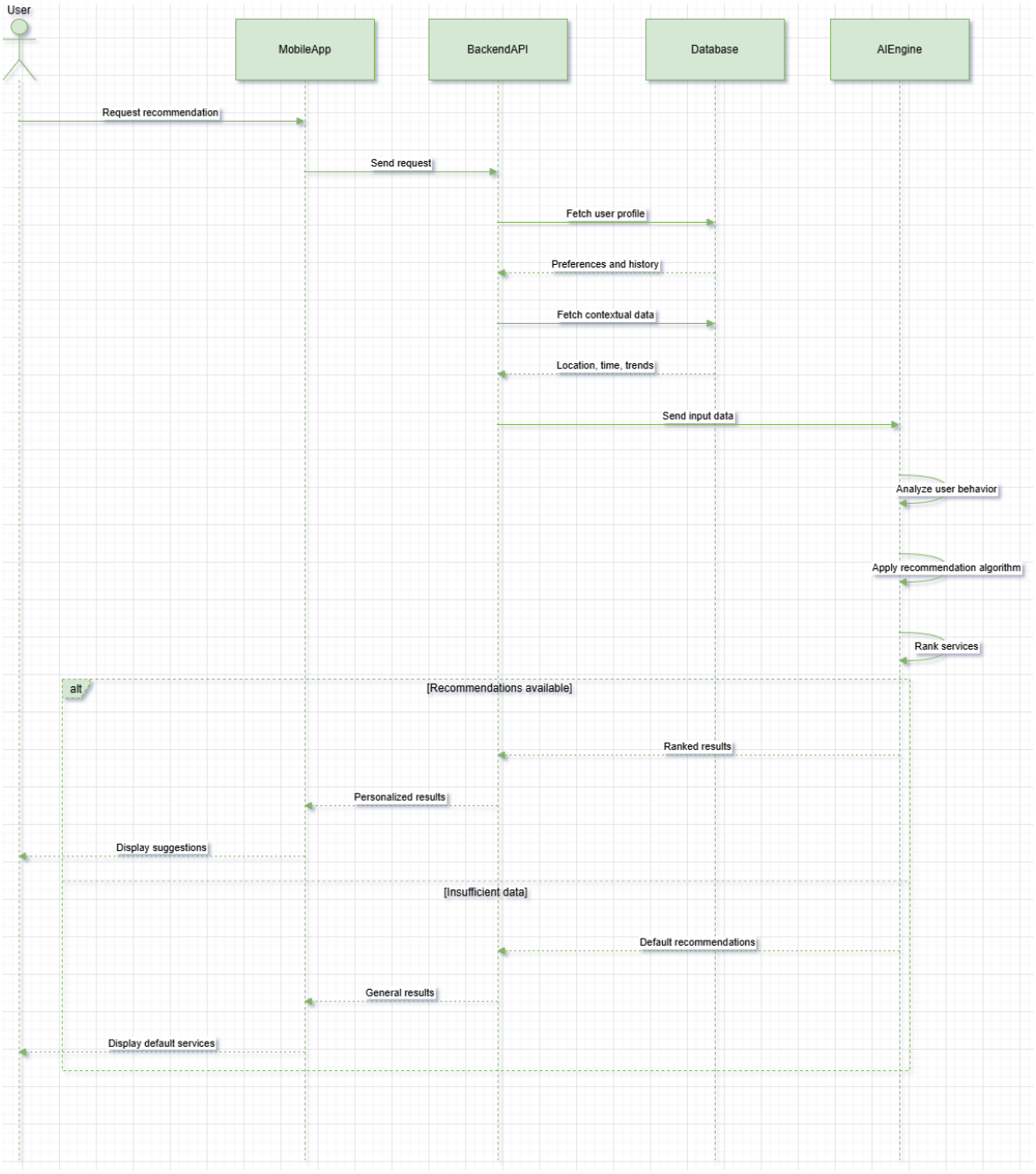


***Figure 26**: AI Assistant Sequence Diagram*

9. Real-Time Tracking Sequence Diagram

This diagram shows how the system provides real-time tracking for services such as rides and deliveries. The user requests tracking information, and the backend continuously retrieves location updates from external APIs. These updates are stored and sent back to the user in real time using a loop structure. This ensures that the user receives live updates about their service status.

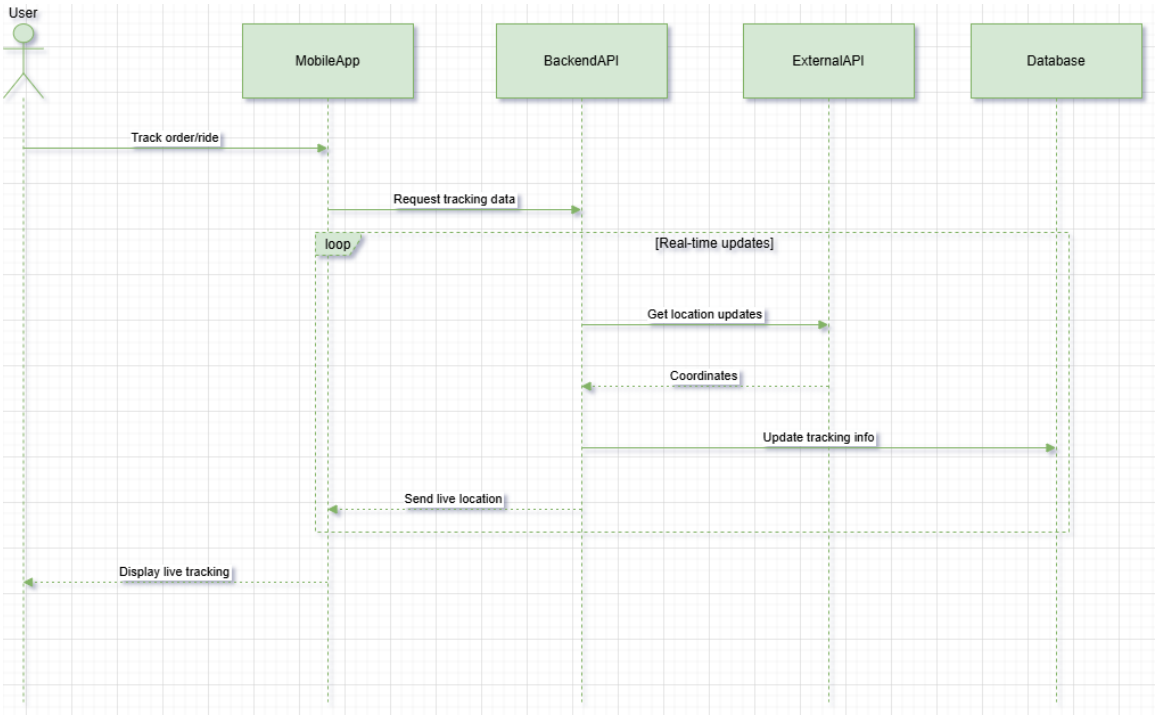


***Figure 27**: Real-Time Tracking Sequence Diagram*

10. Cancellation and Refund Sequence Diagram

This sequence diagram illustrates the process of canceling a booking and issuing a refund. When a user requests cancellation, the backend communicates with the external service provider to cancel the booking. Once confirmed, the system initiates a refund through the payment gateway. The transaction status is updated in the database, and the user is notified of the successful cancellation and refund. This ensures proper handling of exceptional scenarios.

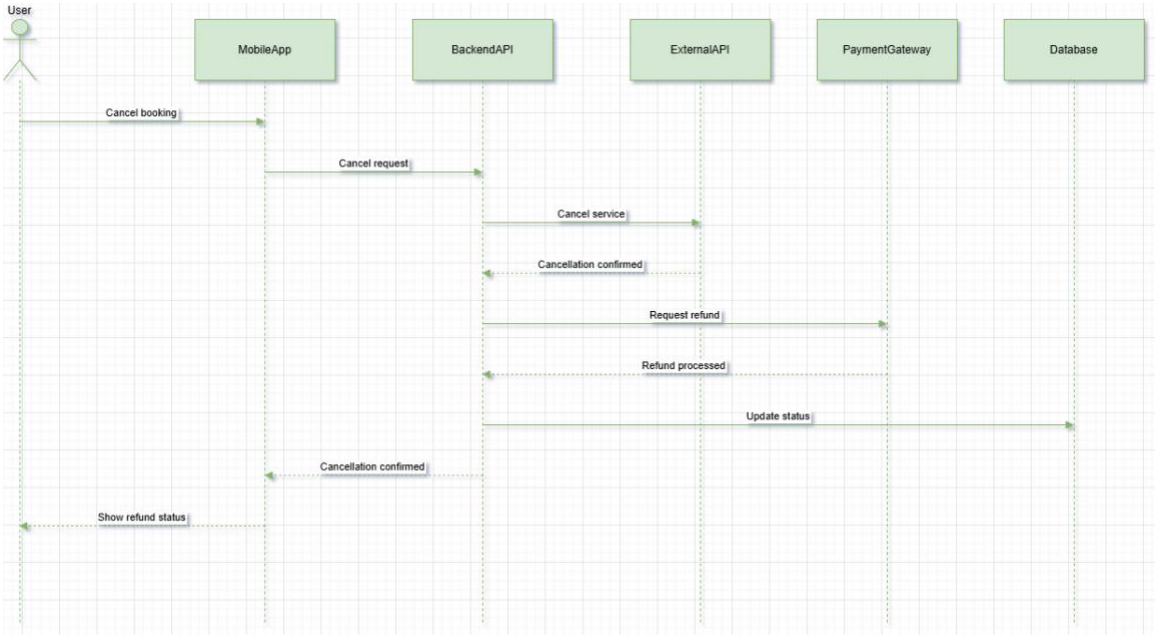


Figure 28: *Cancellation and Refund Sequence Diagram*

3.5.4 Class diagram

This class diagram represents a detailed structural model of the Masarak application, ensuring that all classes include meaningful attributes and behaviors. The design is aligned with the system's use case and sequence diagrams while also reflecting database-oriented entities.

The **User** class manages authentication and interaction with the system and is associated with a **Wallet** for handling financial balance and transactions. The **Service** abstraction is extended into specific service types such as ride, food, hotel, and market, each with its own specialized operations. These services are linked to **Provider** entities to support comparison across multiple platforms.

The **Booking** class serves as the central entity connecting users, services, and providers, and includes operations for creation and cancellation. The **Payment** subsystem supports both wallet-based and external card payments through integration with the **PaymentGateway**, ensuring flexible transaction handling.

The **AIEngine** module processes user-related data through the **UserProfile** class and generates recommendations using the **Recommendation** class, enabling intelligent decision-making. The **Tracking** class provides real-time updates for active bookings, while the **Notification** class manages user alerts triggered by booking and payment events. All classes include relevant attributes and methods, ensuring the design is both functionally complete and suitable for implementation, while maintaining clarity and strong alignment with system behavior.

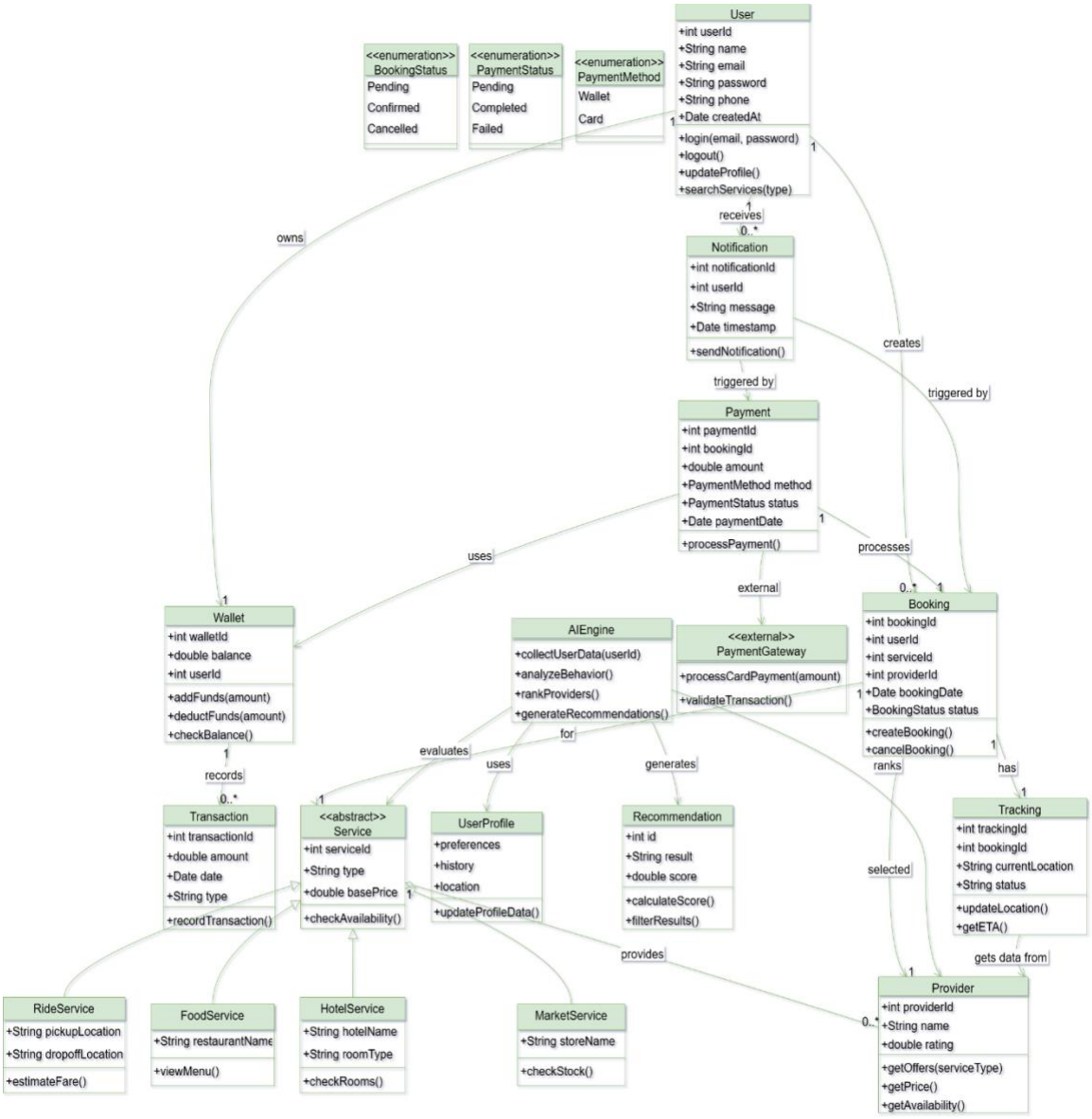


Figure 29: *Class Diagram*

3.6 System Architecture & Design

The overall architecture of the Masarak application is depicted in this section. It shows how each system component interacts with the other so that the user will receive a single efficient experience. The design architecture is that of a client-server architecture, which enables scalability, reliability and secure communications between system components.

3.6.1 System Architecture

Masarak App uses a client-server system by having the mobile app act as the client to the central server, which provides the back-end for the service to accept requests from users, execute the business logic required for the request, and interface with both the database and outside vendors to fulfil these requests.

Supporting elements that provide the interaction between the mobile app and the server are the mobile application, back-end server, back-end server database and third-party providers accessed using APIs.

1. Mobile Application: This is the user interface through which users interact with the system for services such as book rides, order food, and so on.

2. Backend Server: The backend server processes requests, executes application logic, manages user authentication, and provides coordination of communication between the various components.

3. Database: The database provides secure storage for user data, transaction records, and application logic.

4. External APIs (Service Providers): The system is integrated with third-party services, such as ride-sharing services and food delivery services, through APIs. The system accesses and retrieves data from these third-party services, such as service availability, service pricing, etc.

Communication from a user goes through a series of steps in the Mobile Application before being sent to the backend service for processing. The communication starts when the user submits a request using the Mobile Application. After submission, a secure transmission of the user's request is sent by the Mobile Application to the backend server where the server processes the request using appropriate logic related to the request. This processing could either be retrieving information from the Internal Database or issuing a query to external API's to gather needed information. Once the backend server finishes processing the request, it then transmits a response to the mobile application, where the user views the response.

The security and reliability of communication between the system components are accomplished through using encrypted protocols (HTTPS) and implementing the appropriate form of authentication.

The Masarak System is installed on a cloud platform for improved scalability, reliability and high-performance. The nature of Cloud services means all communications between mobile apps, back-end APIs, databases, external service providers and the payment

gateway will be secured with HTTPS/TLS encrypted. The architecture has protection of the API gateway, firewall rules and limited access to internal services.

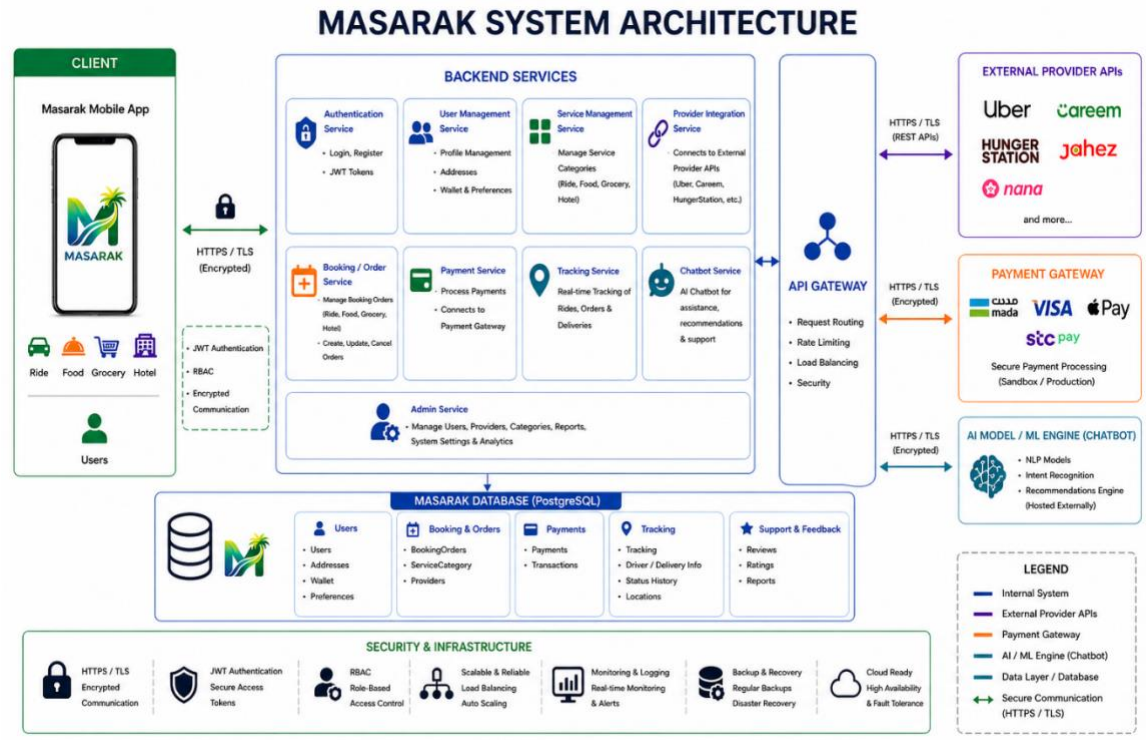


Figure 30: *Masarak System Architecture Diagram*

3.6.2 Security Architecture

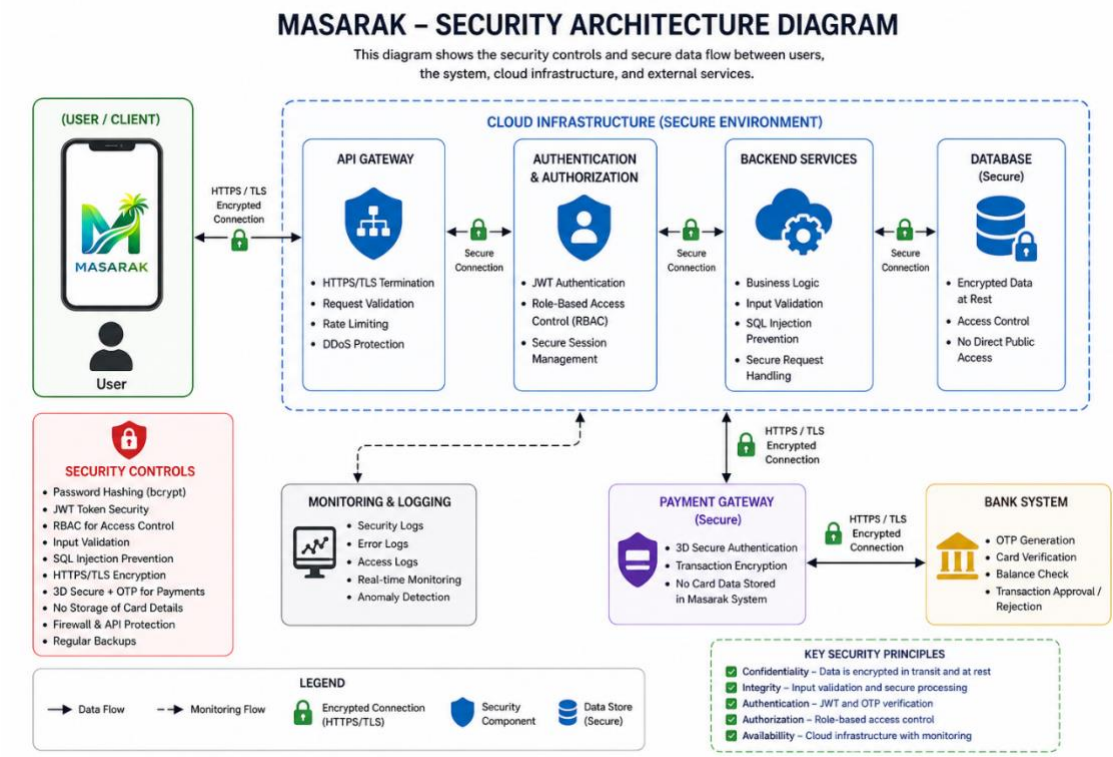


Figure 31: *Masarak Security Architecture Diagram*

The diagram shows the complete Masarak Security Architecture, which defines how all components of the Masarak System will be protected from compromise (criminal hacking attempts) by multiple layers of security within a Cloud Computing Environment.

Through the use of HTTPS/TLS, communication between the Mobile Application and Backend Services, the Mobile Application and Database, and the Mobile Application and External System, is encrypted. This means that all data is transmitted securely and confidentially and prevents third parties from intercepting or accessing data between the mobile application to back/services and/or the database.

All incoming requests are handled by the application programming interface (API) gateway, which also performs request validation and authenticates users to ensure that only authorized users can gain access to the system functions. The authentication and authorization of users are achieved through the use of JSON web tokens (JWT). This process ensures that only users who can be verified will have access to the functions of the system. Role-based access control (RBAC) prevents users from having access to sensitive functions.

The backend services perform input validation and secure processing of data to eliminate many common types of attacks, such as SQL injection. The database has restricted access so that users cannot gain direct access to the database, there are authorized backend services to access the database on behalf of users.

With regard to the processing of payments, the Masarak system uses a secure payment gateway in conjunction with the external bank system to process transactions. The authentication of transactions will be accomplished using the 3D-Secure authentication and OTP validation process prior to being approved in order to provide protection against fraud.

The design of the system uses a layered security structure to provide overall system confidentiality, integrity, and availability through encryption, authentication, and secure handling of data.

3.6.3 Layers

The Masarak Architecture utilizes a layered system architecture where specific responsibilities are defined within each layer, improving maintainability and scalability. Each of the layers has a separate responsibility to help in performing the system's overall functionality.

The layers are as follows:

- Presentation Layer: The presentation layer consists of the mobile application interface and is responsible for displaying information and receiving user input.
- Application (Business Logic) Layer: This layer is responsible for processing user requests, adding business rules to it, and controlling the overall functionality of the system. It also serves as the middleware between the presentation layer and the data layer.
- Data Layer: The data layer is responsible for the management and organization of data in the database. In addition, the data layer handles the retrieval, storage and updating of data while maintaining the integrity and security of the data.

By applying a layered structure for the system architecture, we will gain modular isolation for the various layers of the system architecture, which will make it much easier for us to maintain, upgrade and increase the size of our solution as the user base and services increase exponentially.

3.6.4 Database

- Entity-Relationship Diagram (ERD)

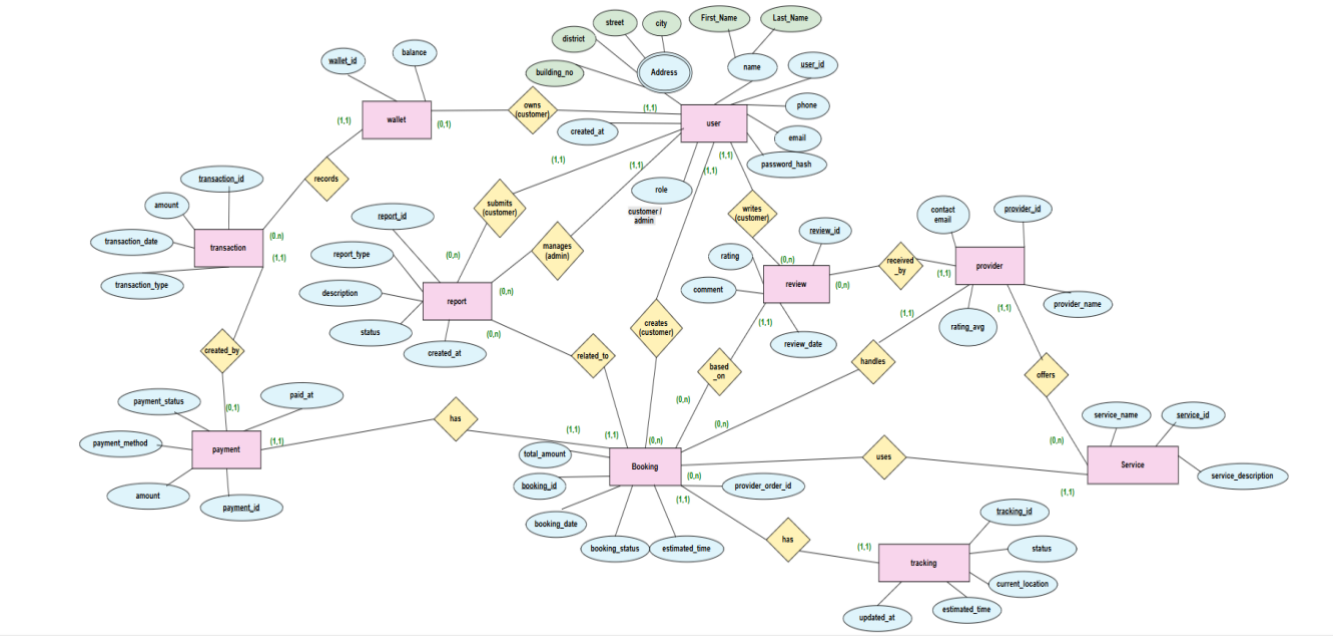


Figure 32: *ER diagram*

- Database Schema

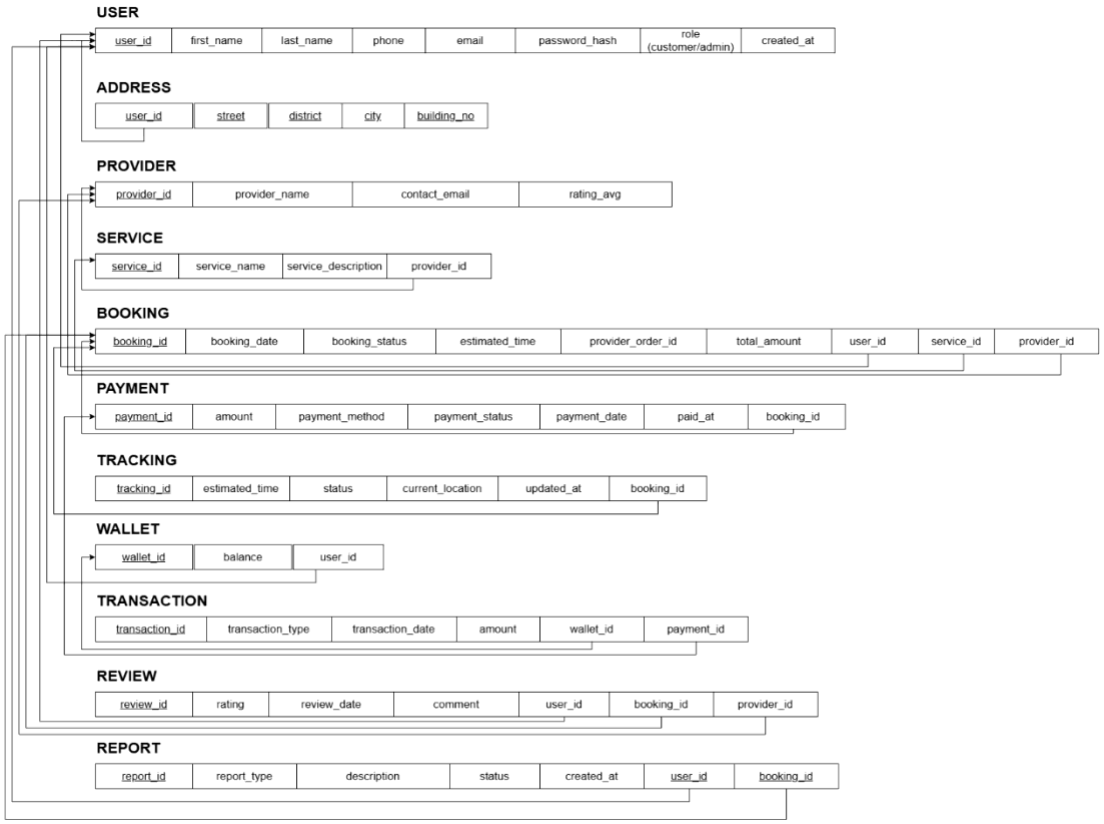


Figure 33: *Database Schema*

Designed as a unified, adaptable database schema, the Masarak system's structure will enable different types of service providers to offer multiple types of services (e.g. food deliveries, ridesharing, online marketplaces) as part of an integrated solution.

The Masarak system is designed so that there will be clear logical separation by service category, while at the same time providing the ability to manage the interrelated relationships among users/providers/ bookings/other related activities and operations using a composite schema model. The composite schema also provides an acceptable way of scaling today and in the future, as well as allowing for a managed method to perform data management tasks.

Through this design, the Masarak database does not contain redundant data while also providing for internal data integrity, and continues to provide for external data exchange via the backend services of the Masarak system. Since the Role Attribute is a part of each user record, the application can identify users with their respective roles for purposes of differentiation based upon their differing roles without requiring separate tables in the database.

Each of the user's addresses is stored as part of the user's data, as one of the attributes of the user entity. This is intended to maintain a simpler database design that will ultimately be easier to maintain over time.

The AI chatbot is not integrated with the main database schema and functions independently. Therefore, it is not necessary for the chatbot to access the database directly for processing user input or generating recommendations by analyzing user habits as well as using data available in the main database, and instead uses backend web services for this purpose. The AI chatbot's database can be set apart from the core database. This will allow you to modularize the architecture of the system and reduce the complexity of the schema.

The Masarak System has an external banking integration via a secure payment gateway. The Masarak System will not store full credit card information or sensitive financial information in its database. However, it will store transaction occurrence information for each transaction including Transaction ID, Amount, Payment Method and Status. The Bank will be responsible for authentication of credit card number(s), OTP validation and balance checking.

The overall design of the database incorporates features such as simplicity, scalability, and maintainability to ensure that it is perfectly suited to meet the functional requirements of the entire application while providing interoperability with other systems.

3.7 Application Design

- Prototype

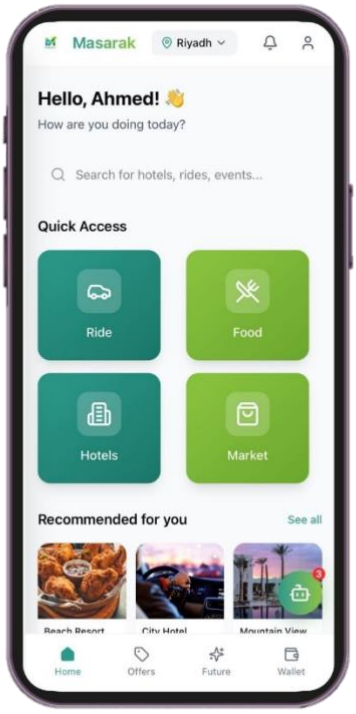


***Figure 34**: Home screen with quick service.*

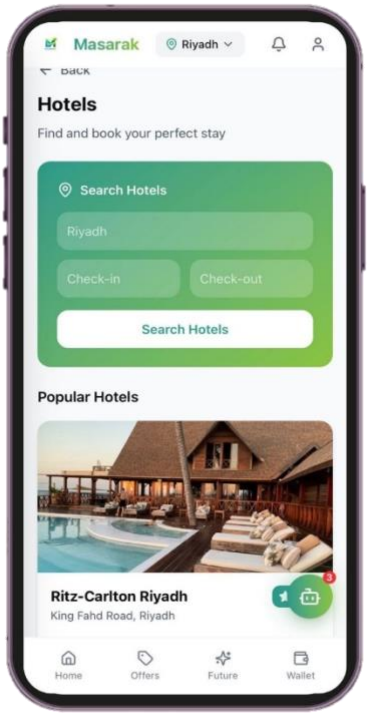


***Figure 35** : Hotel search and booking.*

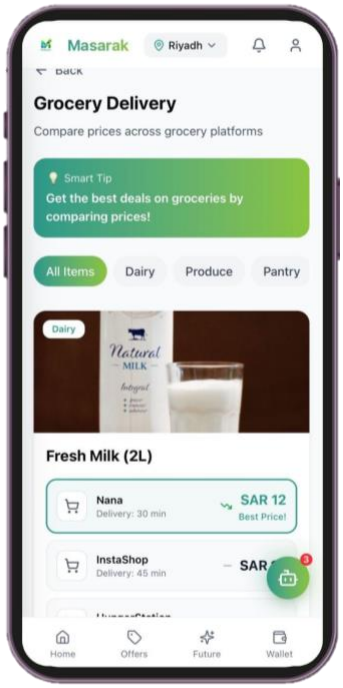


Figure 36 : *Grocery price comparison.*

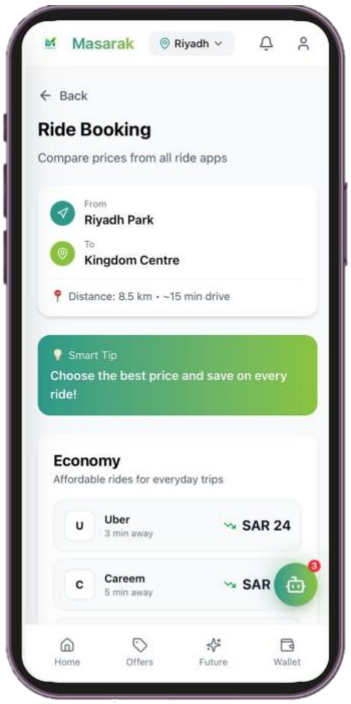


Figure 37 : *Ride price comparison.*

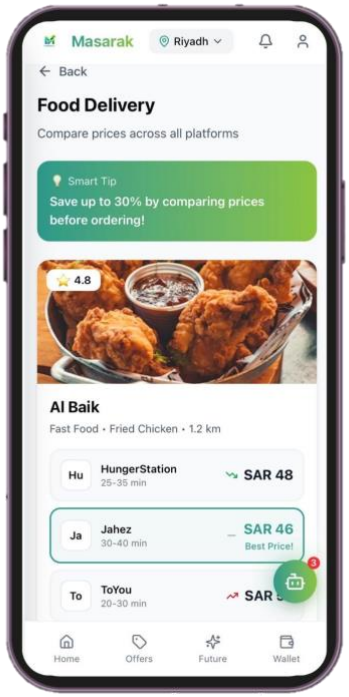


Figure 38 : *Food ordering comparison.*

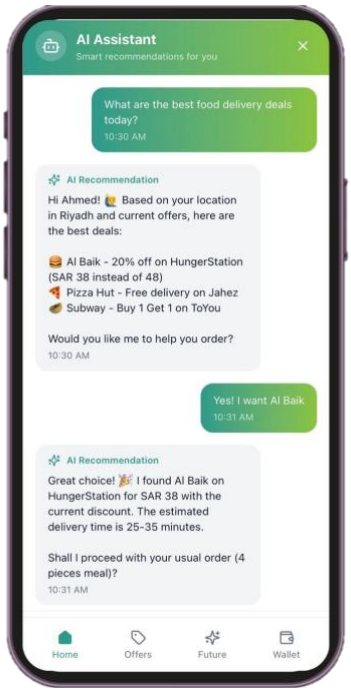


Figure 39 : *AI assistant recommendations.*

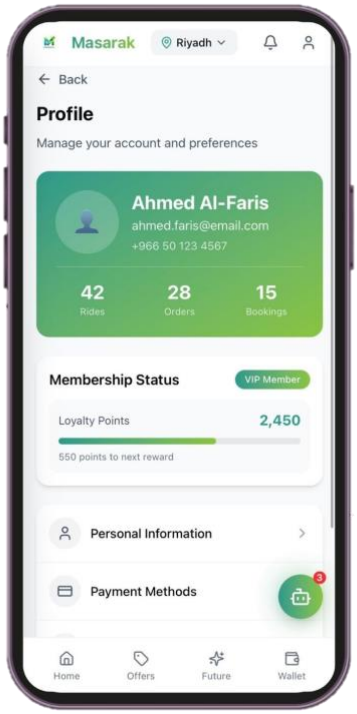


Figure 40 : *User profile overview.*

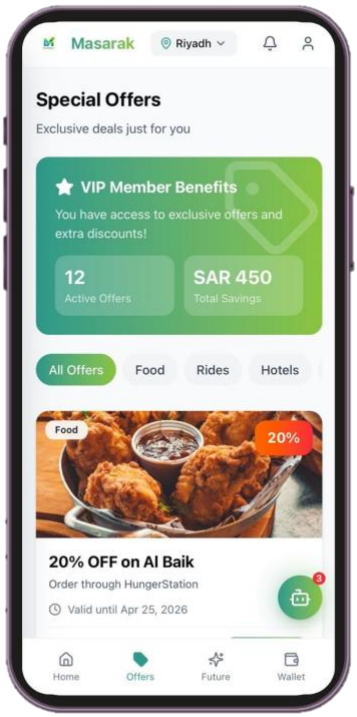


Figure 41 : *Special offers page.*

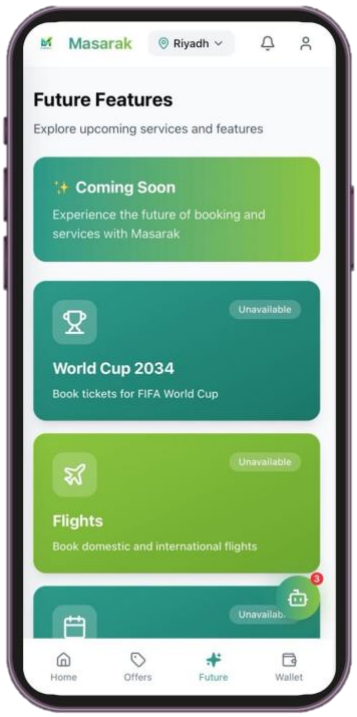


Figure 42 : *Future features section.*



Figure 43 : *World Cup booking page.*

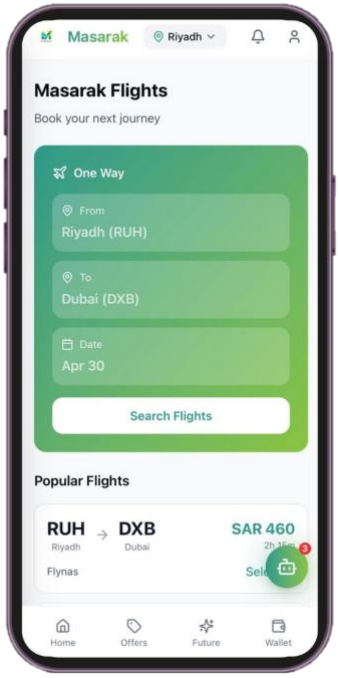


Figure 44 : *Flight booking screen.*

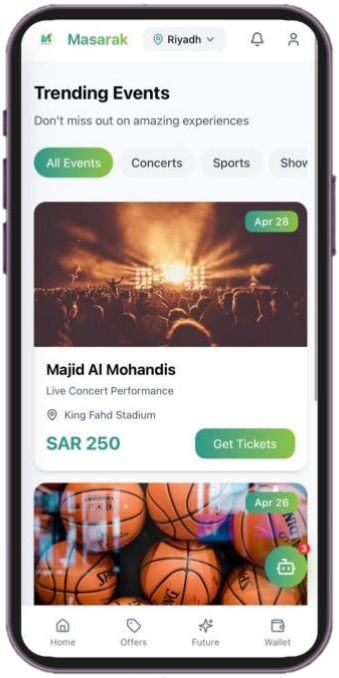


Figure 45 : *Events and ticket booking.*

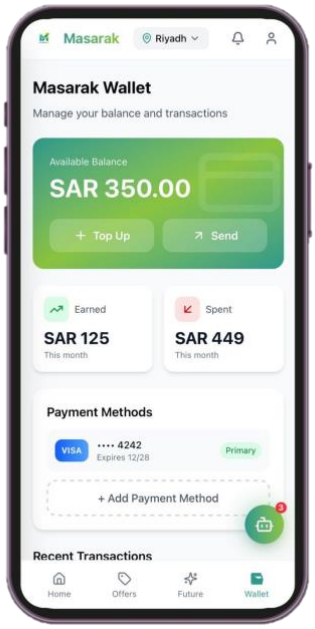


Figure 46 : *Wallet and transactions.*

3.8 System Logic & Behavior

3.8.1 Algorithms / AI logic

Masarak uses an AI-based recommendation system to provide users with personalized suggestions for rides, hotels, food, and grocery services. The recommendation system analyzes user preferences, location, search history, service usage, ratings, and previous orders to generate suitable recommendations. The main purpose of the AI logic is to reduce the time users spend switching between different services and to improve the overall user experience.

The recommendation process begins when the user opens the application or searches for a service. The system collects relevant data such as the user's current location, selected service type, preferred price range, previous bookings, and frequently used services. This data is then processed by the backend system and passed to the recommendation engine.

The recommendation engine applies filtering and ranking logic to identify the most suitable options. For example, if a user usually orders food from nearby restaurants, the system prioritizes restaurants with high ratings, reasonable prices, and fast delivery times. Similarly, for ride services, the system recommends providers based on estimated arrival time, price, and availability.

The AI logic follows these steps:

1. Input Collection: The system receives user data such as location, service request, budget, preferences, and previous activity.
2. Data Filtering: The system removes unavailable, irrelevant, or invalid service options.
3. Personalization: The system compares available services with the user's behavior and preferences.
4. Ranking: The system ranks services based on price, distance, rating, availability, and user preference.
5. Recommendation Output: The system displays the most suitable recommendations to the user.

- A simplified recommendation algorithm can be described as follows:

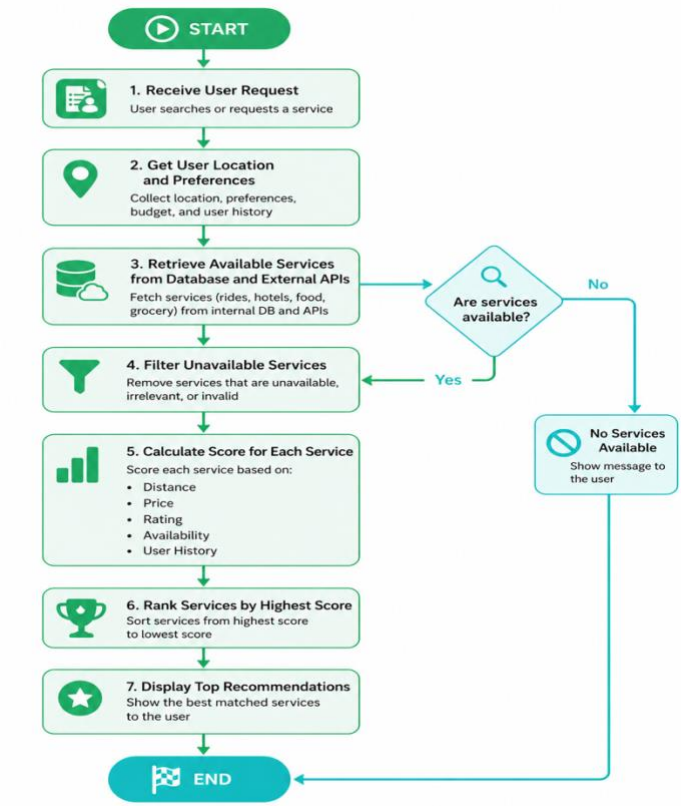


***Figure 47**:AI-Based Recommendation Algorithm Flowchart*

The recommendation score can be calculated using the following formula:

$$Recommendation\ Score = (Location\ Match \times 0.30) + (Price\ Suitability \times 0.25) + (User\ Preference\ Match \times 0.25) + (Service\ Rating \times 0.20)$$

The formula establishes an equilibrium between various elements which need to be evaluated for recommendation choice selection. Users exhibit a stronger preference for local services, so the system gives more weight to their proximity while price and user

choice assist in creating tailored experiences. The service rating system verifies the actual standard of service.

Masarak develops its recommendations through time because it studies the patterns which users display in their behavior. The system will show more accurate results because it develops an ability to change its operation. The system now uses its optimization capabilities to create instant service comparisons which assist users in making their most economical and time-efficient decision. The system determines the most effective option after it evaluates ride providers through a system comparison.

Overall, the AI and algorithm logic in Masarak enhances personalization, supports real-time decision-making, and improves user satisfaction by delivering a smarter and more efficient experience.

Masarak's recommendation system is supported with the included AI assistant to support users in interacting with and getting immediate assistance when needed through a conversational system that allows them to use natural (human-like) language to find services within the application, as well as receive personalized service suggestions.

The AI assistant will be able to help users with their requests regarding finding the lowest-priced transportation from the airport or suggesting restaurants in close proximity that have quick delivery service. Moreover, the AI assistant will guide users through the booking process of their required services while also keeping them updated in real-time, and providing step-by-step assistance.

From a systems perspective, the AI assistant acts as an action-based intelligent agent that continually interacts with the Masarak system by retrieving service feed data collected by service providers, analyzing user-supplied data preferences and matching them with available service options that are being stored within the database. Through this interaction, the AI assistant customizes results based on each user's actions.

This integration allows for real-time comparisons between service providers, allowing the customer to make a better decision as to which service will meet their needs based on the price of the service, the availability of the service and the current needs of the customer. Overall, the AI Assistant increases the usability of the Masarak application, decreases the customer's effort involved in using the Masarak system, and improves the overall efficiency of the Masarak application.

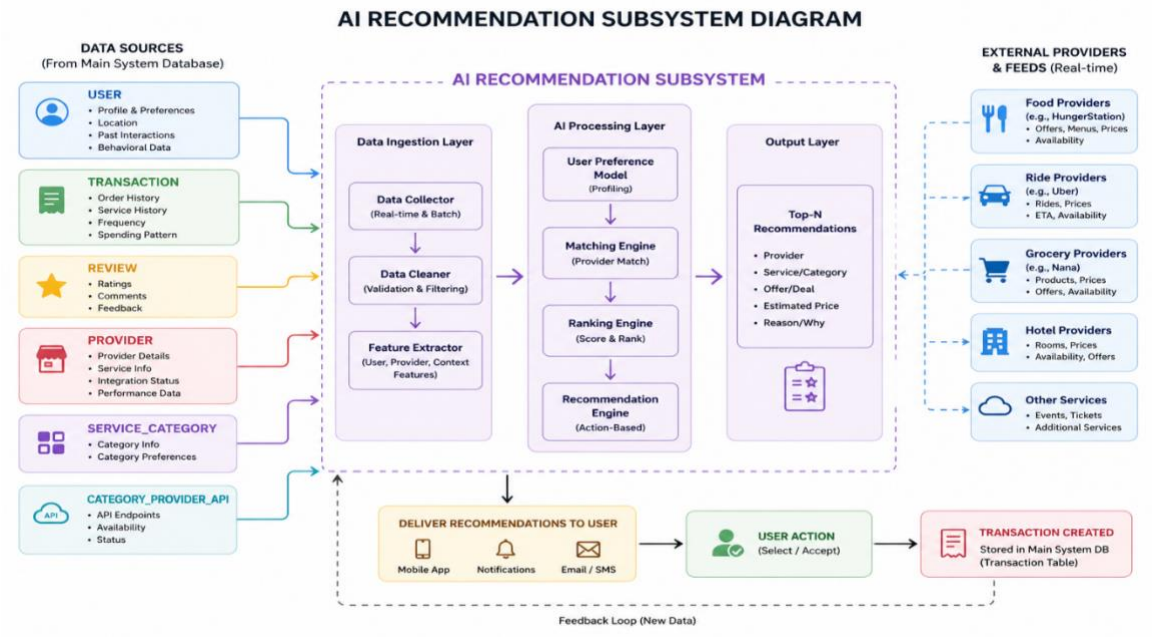


***Figure 48**: AI Recommendation Subsystem Architecture*

3.8.2 Error handling & system responses

To ensure that Masarak is reliable, user-friendly and secure in all scenarios through error handling and response, it is necessary to respond to any errors that occur as the Masarak system depends on user input; the backend processes; database processing; and external API calls.

User errors (i.e. invalid user input, failed to login attempts, unavailable services), possible system interruptions (i.e. network failure, API failures) are detected by the Masarak system so users are provided with meaningful and understandable messages that inform them of the problem without exposing them to any technical information.

When a user attempts to log into the Masarak system, the system presents the user with an invalid user/ password message if their username and/or password does not match the stored credentials. If an external service provider fails to respond to the system's request for the service, the user is registered as having requested the service but it will be temporarily unavailable to the user and will have to be requested again later. If a network error is detected by the system, the user will immediately be notified that connection to the network has been interrupted.

The Masarak system also includes error handling to protect the system from application crashes, ensure data consistency between systems and protect in-process transactions. The Masarak system should validate user information before processing any requests; as well as handle exceptions within the system at the probable point of failure. The Masarak system

is also responsible for ensuring incomplete or unsuccessful processes will not be saved in the database.

Improvements to error handling and response time have increased the confidence users hold for using Masarak; therefore, users will have a more positive experience no matter what types of problems they may face unexpectedly while using the Masarak application.

3.9 Security & Privacy

3.9.1 Privacy Policy

Masarak's privacy policy has been developed because the application contains personal user information such as location and transaction information. It will protect and process whatever personal data or transaction details that Masarak collects in an appropriate and responsible manner.

Masarak will only collect the information necessary to perform its services. For example, user accounts will require identifying information for users, their contact details, service preferences, as well as reservation and order information. This collected data will help improve the user's overall experience, provide a means for fulfilling user requests, and support the overall function of the application.

Secure communication protocols, such as using HTTPS (Hypertext Transfer Protocol Secure) using TLS ,will be utilized to encrypt data while communicating between the

mobile application and the backend server. Additionally, sensitive data should never be available without using a secure way to retrieve it (such as credentials that allow you to access or payment information).

Until an individual requests their own information or data, all system components that have access to that information must keep it safe from access by others. Masarak will not store or sell any of the end-user's information to a third party unless necessary to fulfill the user's request to complete a service (e.g., providing ride booking or food ordering service to an end-user

By providing clear information about the collection and use of user information, and enabling the secure sharing of that information by users, Masarak intends to establish trust with its users and comply with established security and data protection standards.

3.9.2 Security Measures

Masarak implements a set of security mechanisms to protect user data, ensure system integrity, and secure communication between system components. As the system processes sensitive data such as personal information, location data, and financial transactions, appropriate security controls are applied across different layers.

All communication between the mobile application, backend services, and external systems is secured using HTTPS/TLS encryption. This ensures that data is transmitted securely and protected from interception.

The system enforces authentication and authorization using token-based mechanisms. Only authenticated users are allowed to access system features, and Role-Based Access Control (RBAC) is applied to restrict access to sensitive operations based on user roles.

To reduce security risks, the system applies input validation and secure request handling. This helps prevent common attacks such as SQL injection and ensures that user inputs are safely processed.

For payment security, Masarak integrates with a secure payment gateway and external bank system. Card transactions are verified using the 3D Secure protocol, where the user must enter a One-Time Password (OTP). The bank validates the OTP and checks the available balance before approving or rejecting the transaction.

Sensitive data is stored securely, and the system does not store full credit card details. Only necessary transaction information such as transaction ID, amount, and payment status is recorded.

Since the system is deployed on cloud infrastructure, additional security measures such as API gateway protection, firewall rules, and restricted database access are applied. The database is not directly accessible and is only accessed through authorized backend services.

Threat	Example	Mitigation
SQL Injection	Malicious input	Input validation & parameterized queries
Man-in-the-Middle	Intercept payment	HTTPS/TLS encryption
Unauthorized Access	Payment without login	Token-based authentication
Payment Fraud	Stolen card	3D Secure (OTP verification)
Data Exposure	Card leakage	No storage of full card data
Cloud Attack	Direct DB access	Firewall + API Gateway protection

Table 7: *Security Threats and Mitigation Strategies in Masarak System*

3.10 Tools used

- Google Forms

Used to collect user data and feedback through surveys.

- Figma

Used to design the prototype and the user interface design

- Draw.io

Used to create system models and diagrams.

- MySQL

Used to design and implement the EER database model.

3.11 Conclusion

Masarak provides a unified solution for accessing essential daily and travel-related services through a single digital platform. The application achieves better user experience through its integrated system which combines ride services and food delivery and grocery shopping and hotel booking services.

The project focused on designing a system that is efficient, user-friendly, and scalable. The system delivers an intelligent customized experience through its main features which include AI-based recommendations and real-time tracking and secure payment solutions. The system architecture of Chapter 3 established mobile application components and backend service elements and database structure and AI module design to create a dependable system framework.

The developed prototype system proves the solution which was proposed can be successfully implemented in real-world situations. With further development and real-world integration, Masarak has strong potential to evolve into a complete super-app that simplifies how users access digital services.

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APPENDICES

Appendix A: Use Case Description 6

<table>
<tr><td>Use Case ID</td><td colspan="2">UC-06</td></tr>
<tr><td>Use Case Name</td><td colspan="2">Book Hotel</td></tr>
<tr><td>Actors</td><td colspan="2">• Primary: User
• Supporting: Hotel Provider</td></tr>
<tr><td>Description</td><td colspan="2">Allow users to reserve hotel rooms.</td></tr>
<tr><td>Preconditions</td><td colspan="2">• Logged in</td></tr>
<tr><td>Main Flow</td><td>Actor Action</td><td>System response</td></tr>
<tr><td></td><td>1. Select hotel

2. Choose room

3. Confirm</td><td>1. Show rooms

2. Calculate price

3. Create booking</td></tr>
<tr><td>Postconditions</td><td colspan="2">• Reservation confirmed</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">• Change dates</td></tr>
<tr><td>Exception Flows</td><td colspan="2">• No rooms available</td></tr>
<tr><td>Dependency</td><td colspan="2">• Compare Prices, Payment</td></tr>
<tr><td>Assumptions</td><td colspan="2">• Availability is accurate</td></tr>
</table>

Appendix 1: *Use Case Description 6: Book Hotel*

Appendix B: Use case description 7

<table>
<tr><td>Use Case ID</td><td colspan="2">UC-07</td></tr>
<tr><td>Use Case Name</td><td colspan="2">Make Payment</td></tr>
<tr><td>Actors</td><td colspan="2">• Primary: User
• Supporting: Payment Gateway</td></tr>
<tr><td>Description</td><td colspan="2">Processes payment transactions.</td></tr>
<tr><td>Preconditions</td><td colspan="2">• Booking/order exists</td></tr>
<tr><td>Main Flow</td><td>Actor Action</td><td>System response</td></tr>
<tr><td></td><td>1. Select method

2. Enter details

3. Confirm</td><td>1. Connect gateway

2. Validate

3. Complete payment</td></tr>
<tr><td>Postconditions</td><td colspan="2">• Payment successful</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">• Cancel payment</td></tr>
<tr><td>Exception Flows</td><td colspan="2">• Payment failed</td></tr>
<tr><td>Dependency</td><td colspan="2">• External Gateway</td></tr>
<tr><td>Assumptions</td><td colspan="2">• Secure connection</td></tr>
</table>

Appendix 1: *Use Case Description 7: Make Payment*

Appendix C: Use case description 8

<table>
<tr><td>Use Case ID</td><td colspan="2">UC-08</td></tr>
<tr><td>Use Case Name</td><td colspan="2">Manage User Reports</td></tr>
<tr><td>Actors</td><td colspan="2">• Primary: Admin
• Admin reviews and handles user reports.</td></tr>
<tr><td>Description</td><td colspan="2">Allows a new user to create an account in the Masarak System.</td></tr>
<tr><td>Preconditions</td><td colspan="2">• Admin logged in
• Reports exist</td></tr>
<tr><td>Main Flow</td><td>Actor Action</td><td>System response</td></tr>
<tr><td></td><td>1. Open reports

2. Select report

3. Take action</td><td>1. Display list

2. Show details

3. Update status</td></tr>
<tr><td>Postconditions</td><td colspan="2">• Report resolved/updated</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">• Filter reports</td></tr>
<tr><td>Exception Flows</td><td colspan="2">• Data error</td></tr>
<tr><td>Dependency</td><td colspan="2">• Report system</td></tr>
<tr><td>Assumptions</td><td colspan="2">• Reports are valid</td></tr>
</table>

Appendix 2: *Use Case Description 8: Manage User Reports*

Appendix D: Use case description 9

<table>
<tr><td>Use Case ID</td><td colspan="2">UC-09</td></tr>
<tr><td>Use Case Name</td><td colspan="2">Track Services</td></tr>
<tr><td>Actors</td><td colspan="2">• Primary: User</td></tr>
<tr><td>Description</td><td colspan="2">Allows users to track their bookings and services.</td></tr>
<tr><td>Preconditions</td><td colspan="2">• User logged in</td></tr>
<tr><td>Main Flow</td><td>Actor Action</td><td>System response</td></tr>
<tr><td></td><td>1. Open tracking
2. View status</td><td>1. Fetch data
2. Display updates</td></tr>
<tr><td>Postconditions</td><td colspan="2">• Status shown</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">• Refresh</td></tr>
<tr><td>Exception Flows</td><td colspan="2">• No data</td></tr>
<tr><td>Dependency</td><td colspan="2">• Booking system</td></tr>
<tr><td>Assumptions</td><td colspan="2">• Data is updated</td></tr>
</table>

Appendix 3: *Use Case Description 9: Track Services*

Appendix E: Use case description 10

<table>
<tr><td>Use Case ID</td><td colspan="2">UC-10</td></tr>
<tr><td>Use Case Name</td><td colspan="2">Manage Wallet</td></tr>
<tr><td>Actors</td><td colspan="2">• Primary: User</td></tr>
<tr><td>Description</td><td colspan="2">Allows users to manage their wallet balance.</td></tr>
<tr><td>Preconditions</td><td colspan="2">• User logged in</td></tr>
<tr><td>Main Flow</td><td>Actor Action</td><td>System response</td></tr>
<tr><td></td><td>1. Open wallet

2. Add/use funds</td><td>1. Show balance

3. Update balance</td></tr>
<tr><td>Postconditions</td><td colspan="2">• Wallet updated</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">• Cancel transaction</td></tr>
<tr><td>Exception Flows</td><td colspan="2">• Payment failure</td></tr>
<tr><td>Dependency</td><td colspan="2">• Payment</td></tr>
<tr><td>Assumptions</td><td colspan="2">• Transactions valid</td></tr>
</table>

Appendix 4: *Use Case Description 10: Manage Wallet*

Appendix F: Use case description 11

<table>
<tr><td>Use Case ID</td><td colspan="2">UC-11</td></tr>
<tr><td>Use Case Name</td><td colspan="2">AI Recommendations</td></tr>
<tr><td>Actors</td><td colspan="2">• Primary: User</td></tr>
<tr><td>Description</td><td colspan="2">Provides personalized service suggestions.</td></tr>
<tr><td>Preconditions</td><td colspan="2">• User logged in</td></tr>
<tr><td>Main Flow</td><td>Actor Action</td><td>System response</td></tr>
<tr><td></td><td>1 Request recommendations

2. View suggestions</td><td>1. Analyze data

2. Display results</td></tr>
<tr><td>Postconditions</td><td colspan="2">• Recommendations shown</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">• Refine preferences</td></tr>
<tr><td>Exception Flows</td><td colspan="2">• No data</td></tr>
<tr><td>Dependency</td><td colspan="2">• AI module</td></tr>
<tr><td>Assumptions</td><td colspan="2">• Data is accurate</td></tr>
</table>

Appendix 5: *Use Case Description 11: AI Recommendations*

Appendix G: Use case description 12

Use Case ID	**UC-12**	
Use Case Name	Manage Users	
Actors	• Primary: Admin	
Description	Allows admin to manage system users.	
Preconditions	• Admin logged in	
Main Flow	**Actor Action**	**System response**
	1. View users 2. Edit/delete	1. Display list 2. Update system
Postconditions	• User data updated	
Alternative Flow	• Search/filter	
Exception Flows	• System error	
Dependency	• Admin access	
Assumptions	• Admin authorized	

Appendix 6: *Use Case Description 12: Manage Users*

Appendix H: Use Case Description 13

<table>
<tr><td>Use Case ID</td><td>UC-13</td></tr>
<tr><td>Use Case Name</td><td>Manage Providers</td></tr>
<tr><td>Actors</td><td>• Primary: Admin</td></tr>
<tr><td>Description</td><td>Allows admin to manage service providers.</td></tr>
<tr><td>Preconditions</td><td>• Admin logged in</td></tr>
<tr><td>Main Flow</td><td><table><tr><th>Actor Action</th><th>System response</th></tr><tr><td>1. View providers

2. Update/remove</td><td>1. Show list

2. Save changes</td></tr></table></td></tr>
<tr><td>Postconditions</td><td>• Providers updated</td></tr>
<tr><td>Alternative Flow</td><td>• Filter</td></tr>
<tr><td>Exception Flows</td><td>• Error</td></tr>
<tr><td>Dependency</td><td>• Provider system</td></tr>
<tr><td>Assumptions</td><td>• Data valid</td></tr>
</table>

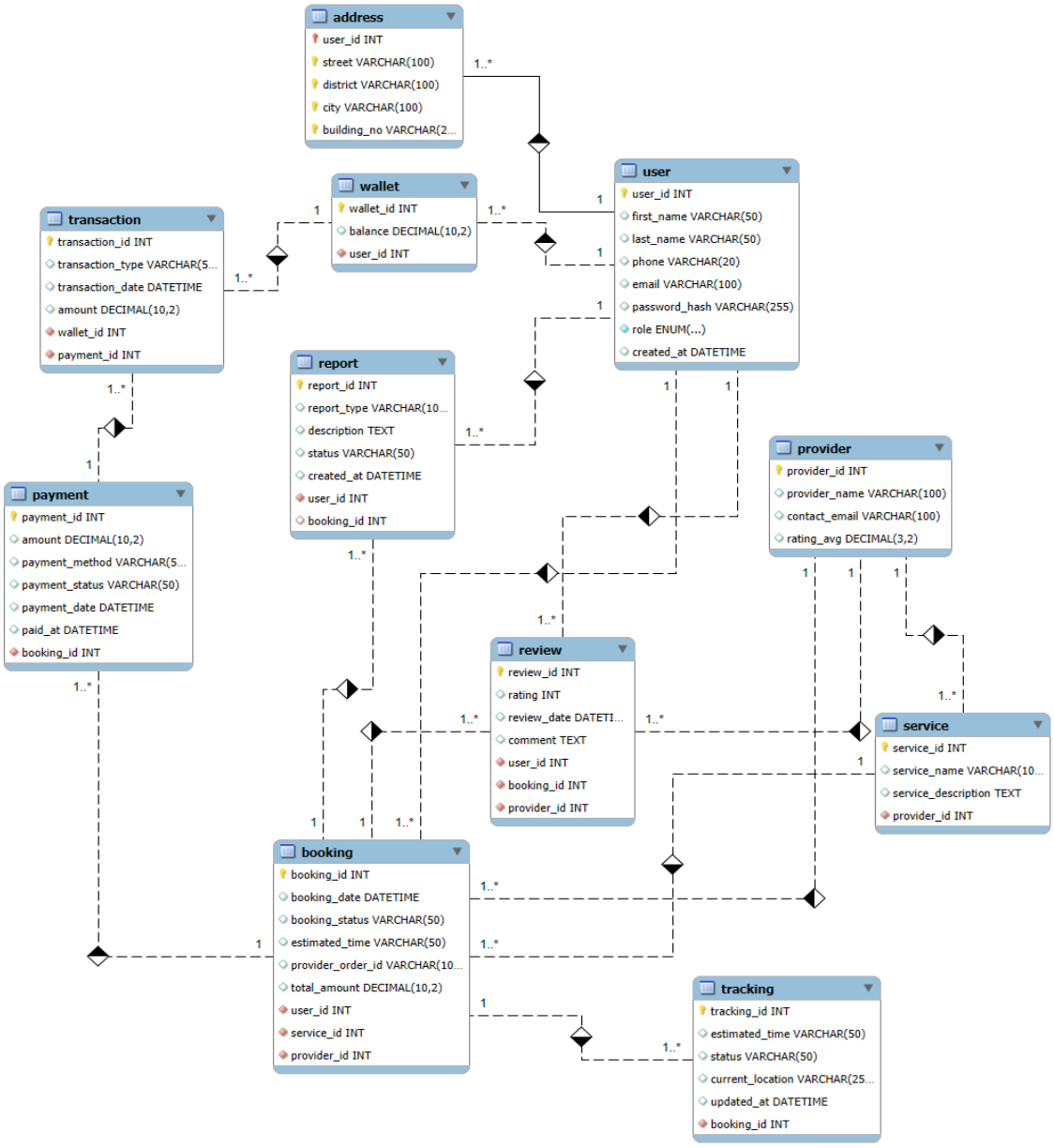
Appendix 7: *Use Case Description 13: Manage providers*

Appendix I: Use case description 14

<table>
<tr><td>Use Case ID</td><td colspan="2">UC-14</td></tr>
<tr><td>Use Case Name</td><td colspan="2">Manage User Reports</td></tr>
<tr><td>Actors</td><td colspan="2">• Primary: System Admin</td></tr>
<tr><td>Description</td><td colspan="2">Allows admin to review and handle user reports.</td></tr>
<tr><td>Preconditions</td><td colspan="2">• Admin logged in</td></tr>
<tr><td>Main Flow</td><td>Actor Action</td><td>System response</td></tr>
<tr><td></td><td>1. Open reports
2. Select report
3. Take action</td><td>1. Display list
2. Show details
3. Update status</td></tr>
<tr><td>Postconditions</td><td colspan="2">• Reports resolved</td></tr>
<tr><td>Alternative Flow</td><td colspan="2">• Filter/add notes</td></tr>
<tr><td>Exception Flows</td><td colspan="2">• Load/update error</td></tr>
<tr><td>Dependency</td><td colspan="2">• Report system</td></tr>
<tr><td>Assumptions</td><td colspan="2">• Reports are valid</td></tr>
</table>

Appendix 8: *Use Case Description 14: Manage User Reports*

Appendix J : Relational Model



Appendix 9: *Relational Model*